

Ryan Camilleri,
Jason Lee, &
Tamara Davis

The University of Queensland

On behalf of the Unite team!

SN-UNITE!

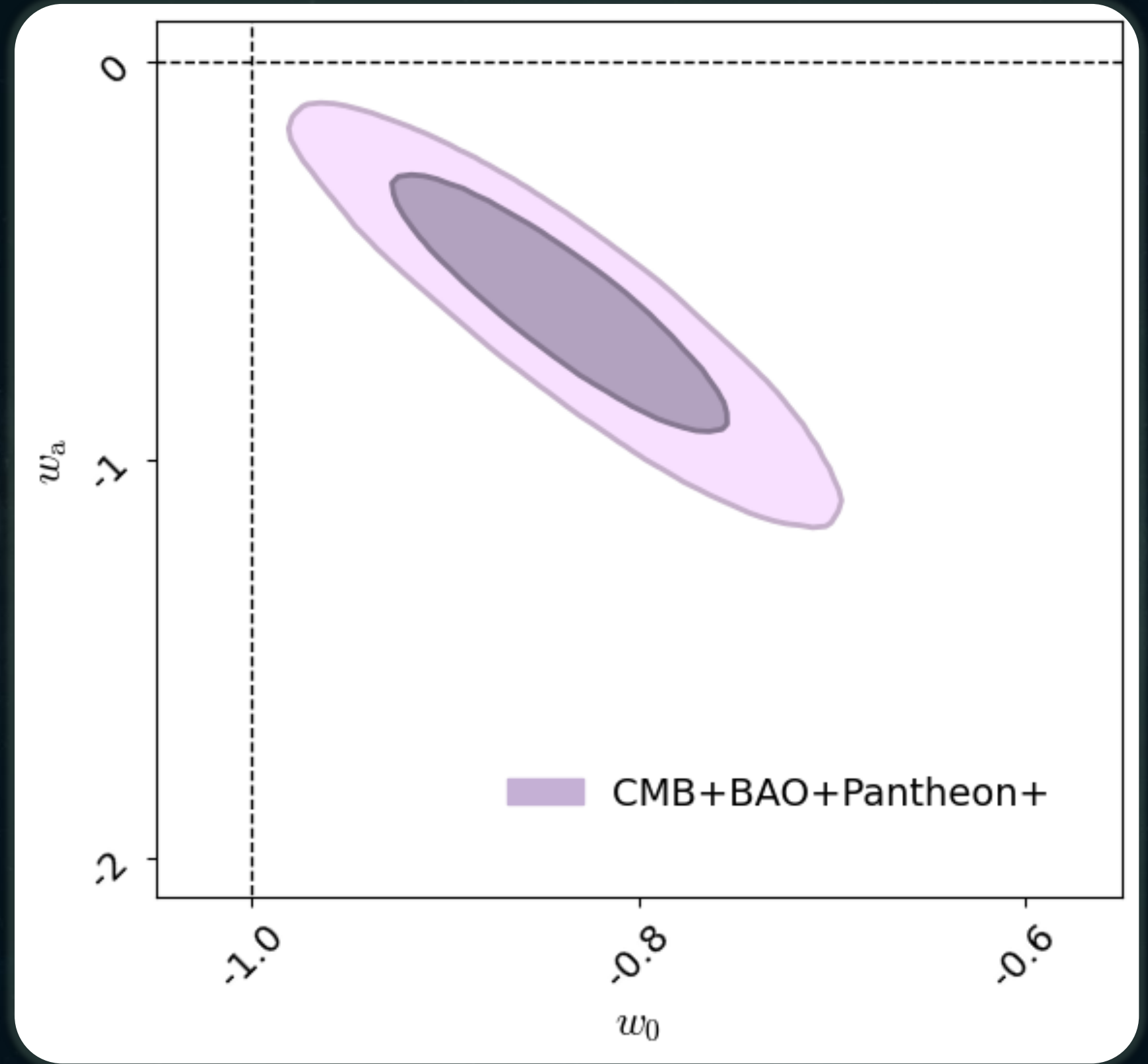
Camilleri et al: 2609.05053
Lee et al.: 2609.05321

Data URL coming soon!

SN-UNITE SUMMARY

Unite! The largest supernova compilation to date. ~2900 type Ia supernovae give tight constraints on dynamic dark energy

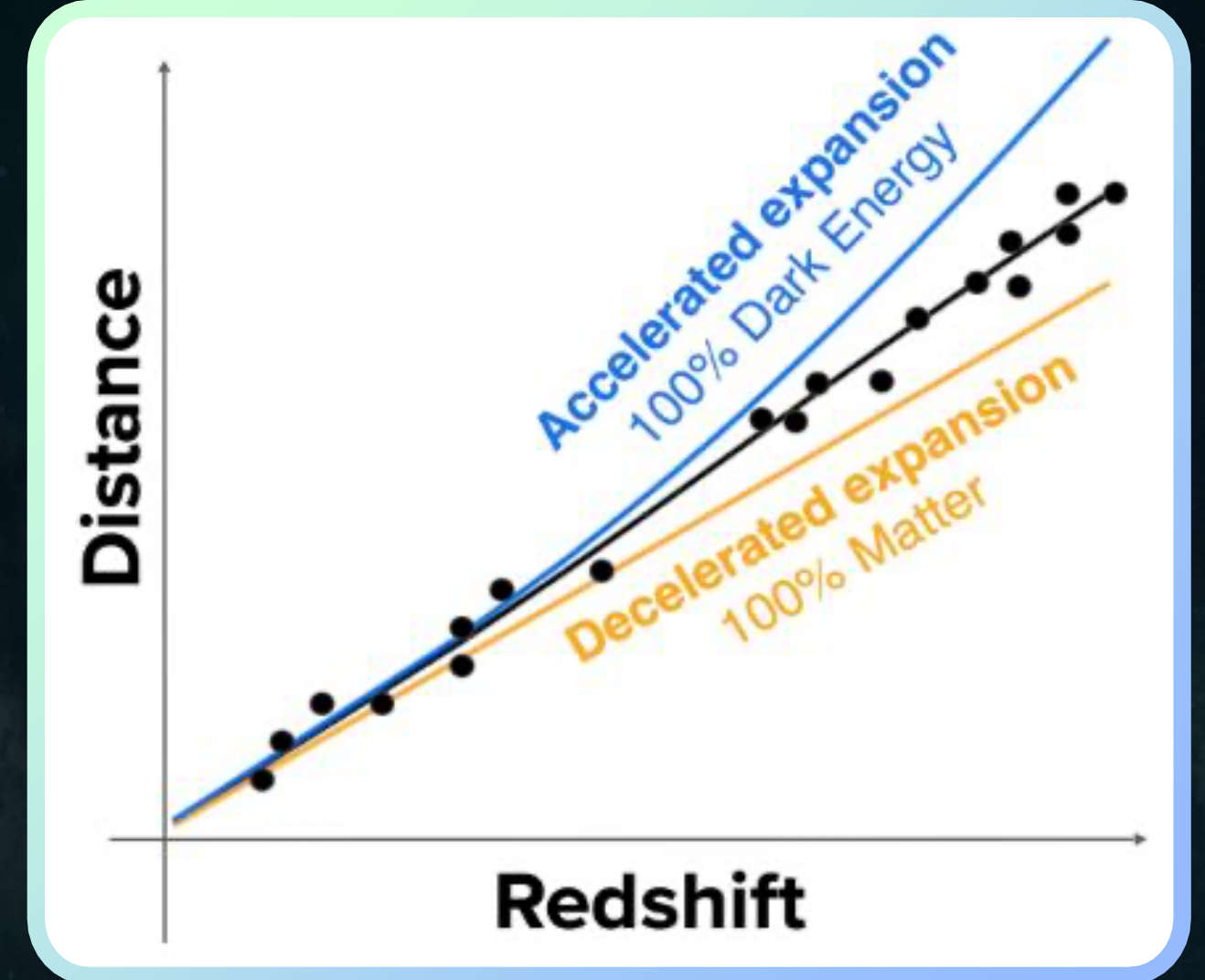
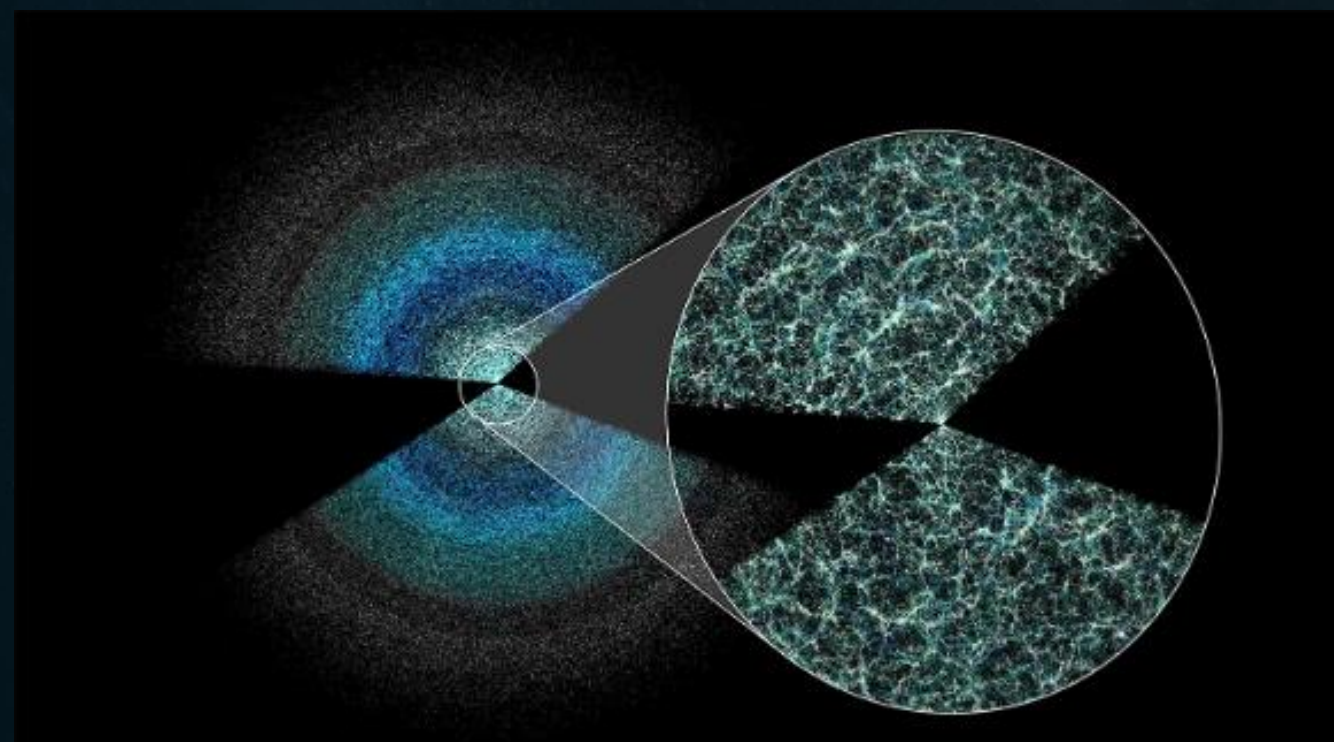
Upgrades galore! Consistently combines the two largest supernova compilations, bringing the latest updates to Pantheon+, including new host galaxy masses



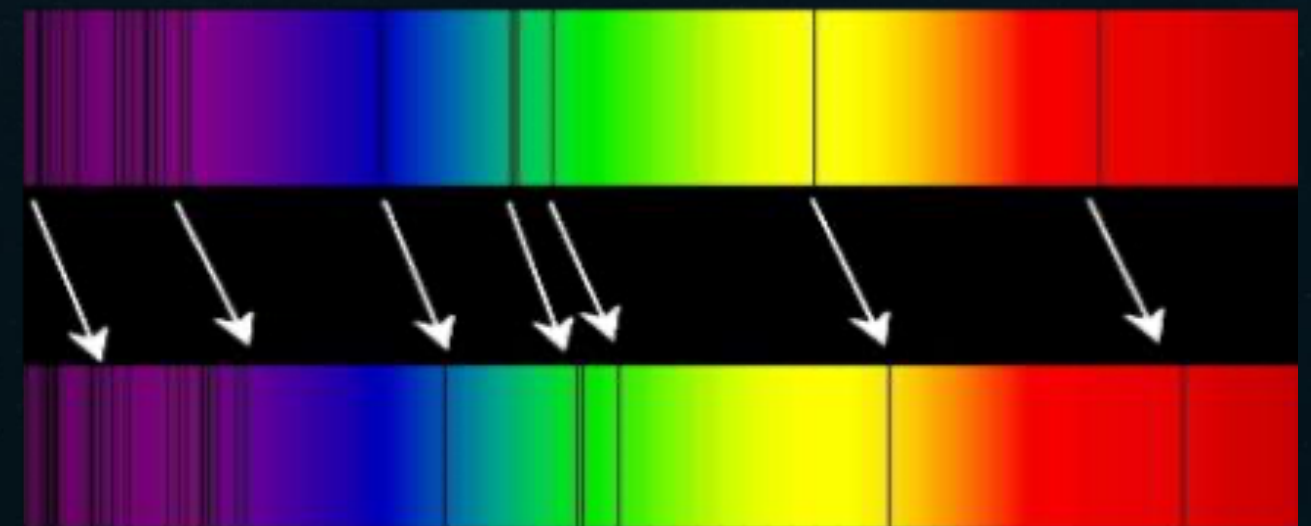
Hubble Diagram

COSMOLOGY

01. Measure Distance



02. Measure Redshift



Hubble Diagram

COSMOLOGY

$$\text{Distance } D = c \int_0^z \frac{dz}{H(z)}$$

Speed of light

Redshift

Hubble parameter

$$H(z) = H_0 \sum_i \Omega_i (1+z)^{3(1+w_i)}$$

Hubble constant

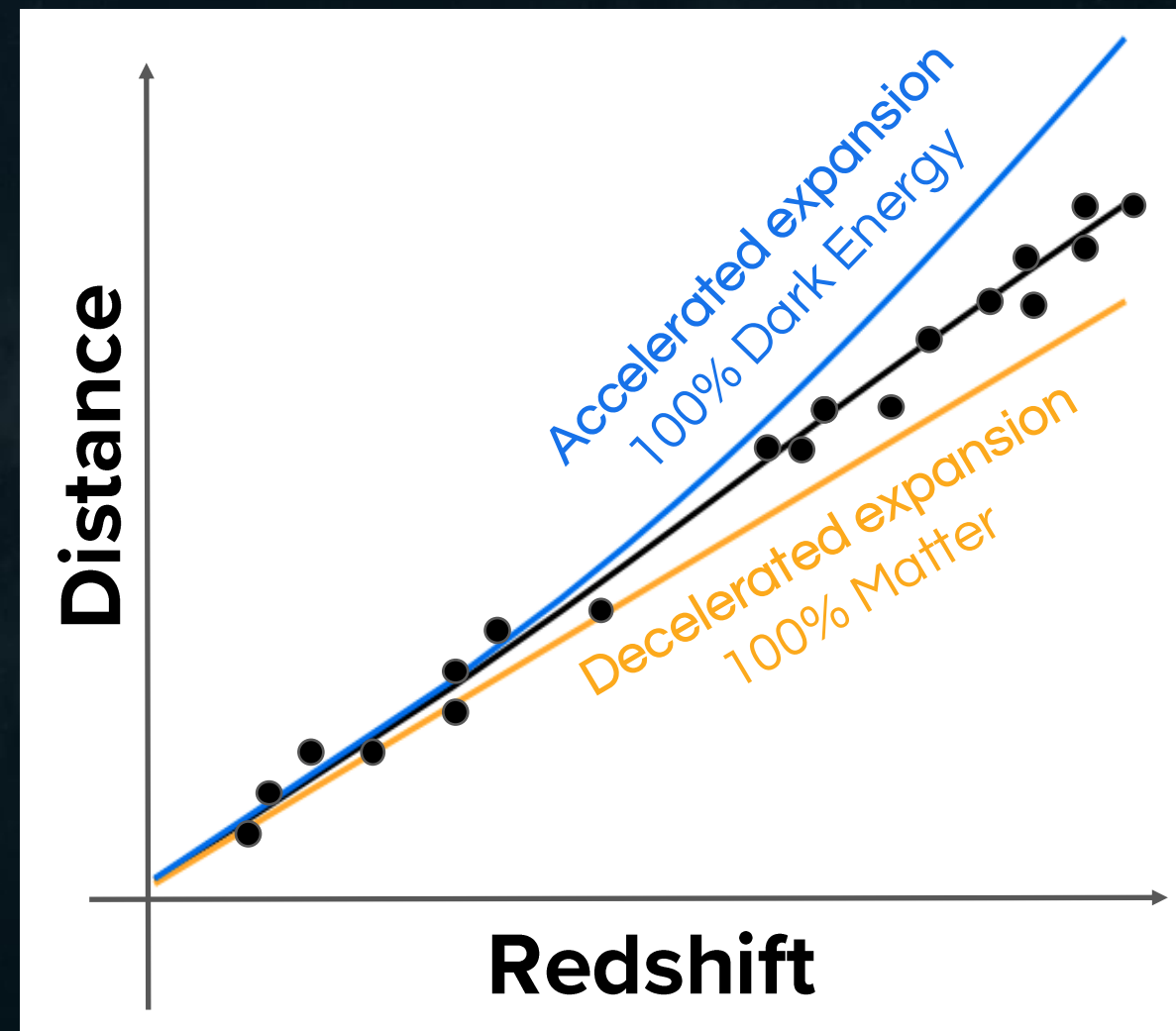
Density

Equation of state

Matter: $w_M=0$

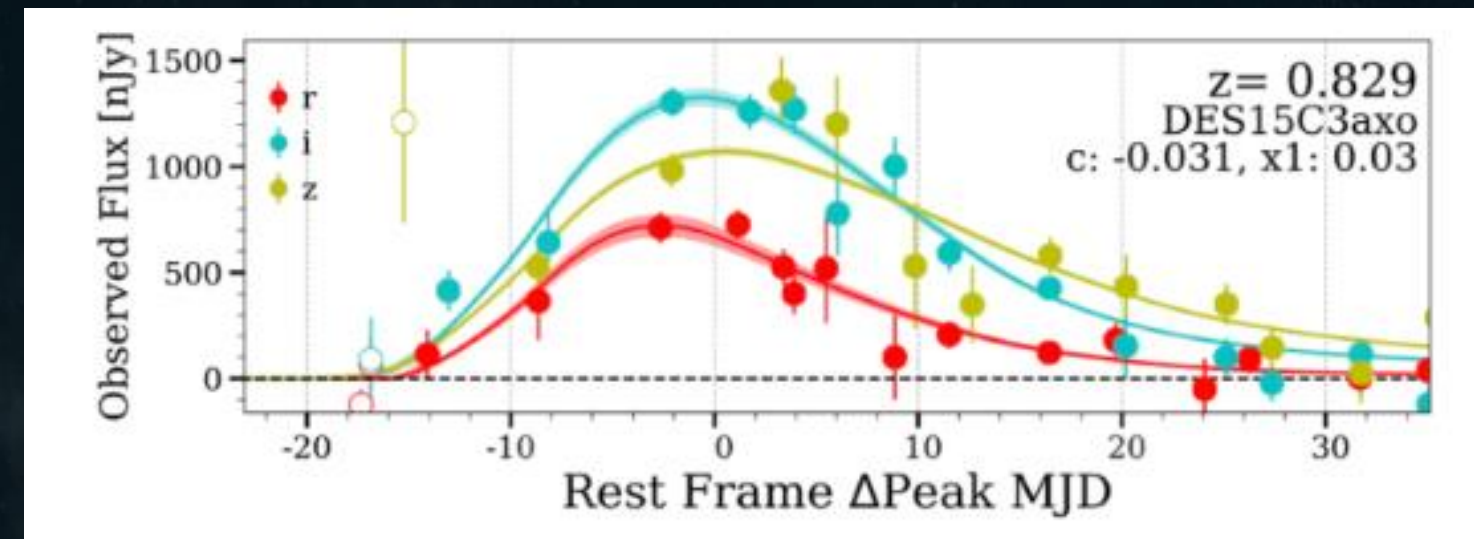
Dark Energy (cosmological constant): $w_\Lambda=-1$

Dark Energy (dynamic): $w(\alpha) = w_0 + w_\alpha(1-\alpha)$





SUPERNOVAE



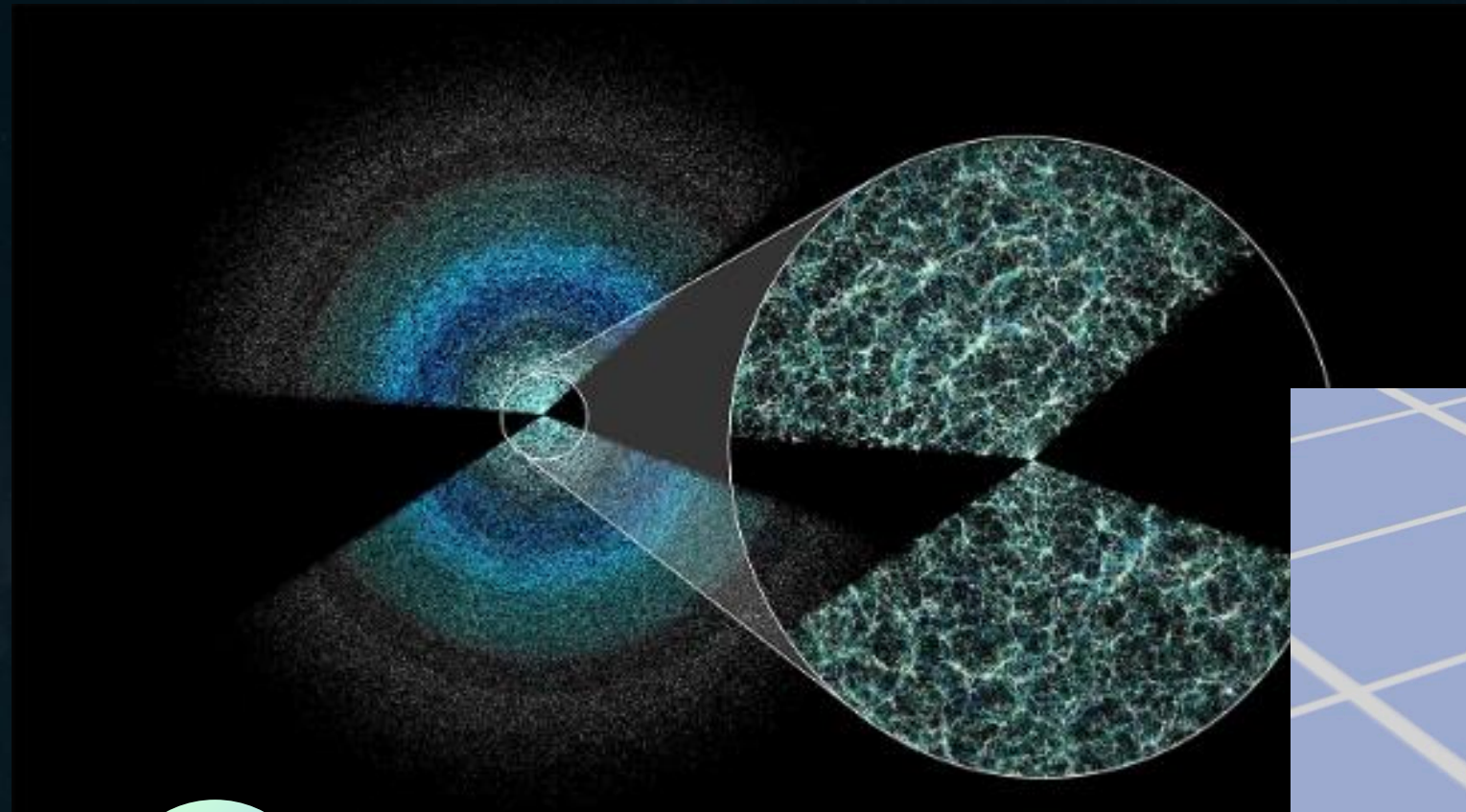
1

Estimate Peak Magnitude

$$\begin{array}{ccccccccccc}
 \text{apparent} & & \text{Measured from each} & & & & & & \text{weak} & & \text{Measured from each} \\
 \text{magnitude} & & \text{supernova light curve} & & & & & & \text{lensing} & & \text{host and l.o.s.} \\
 & & & & \text{stretch} & \text{colour} & \text{host mass} & \text{correction} & & & \\
 \mu_B = m_B^* - M_B + \alpha x_1 - \beta C + \gamma G_{\text{host}} - \Delta\mu_{\text{lens}} + \Delta\mu_{\text{bias}} \\
 \text{absolute} & \text{stretch} & \text{colour} & \text{mass} & \text{Measured for the} & \text{Measured for the} \\
 \text{magnitude} & \text{scale} & \text{scale} & \text{scale} & \text{population} & \text{specific data set}
 \end{array}$$

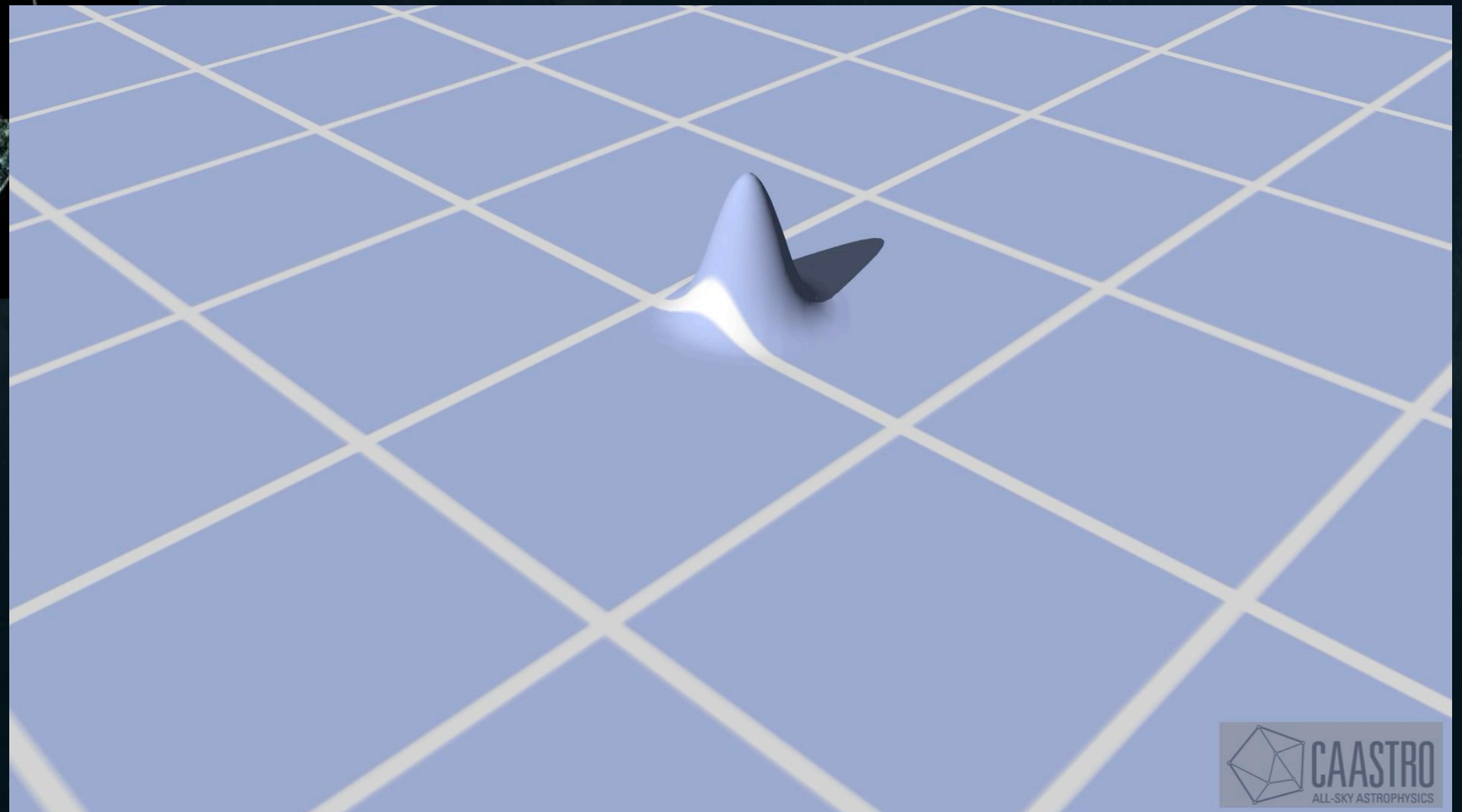
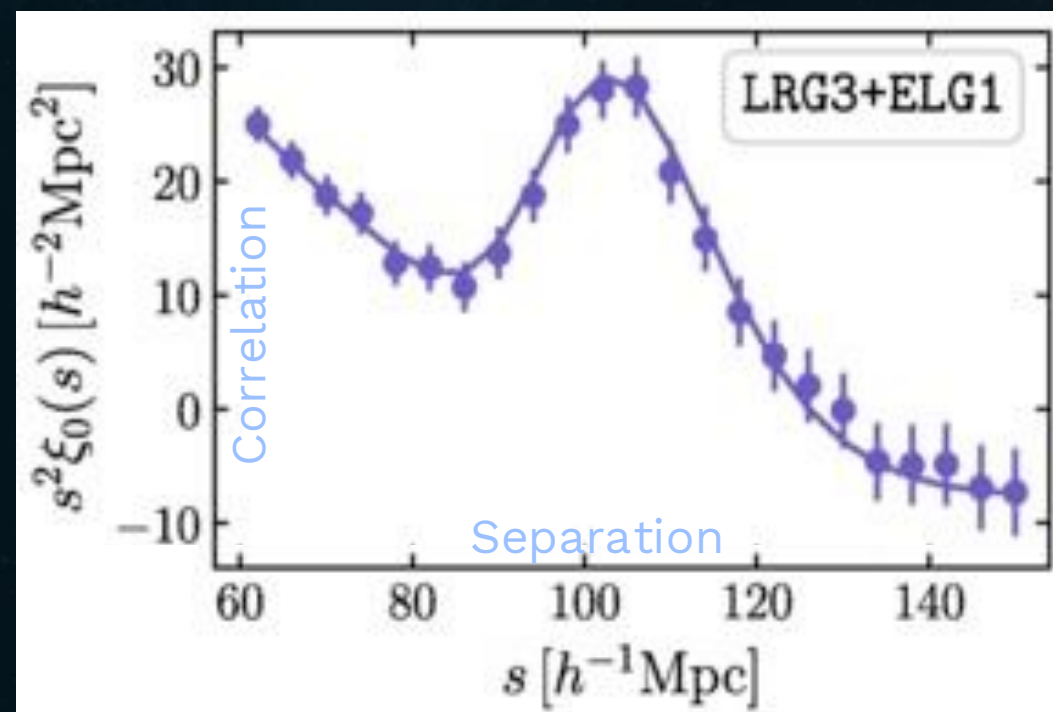
BAO

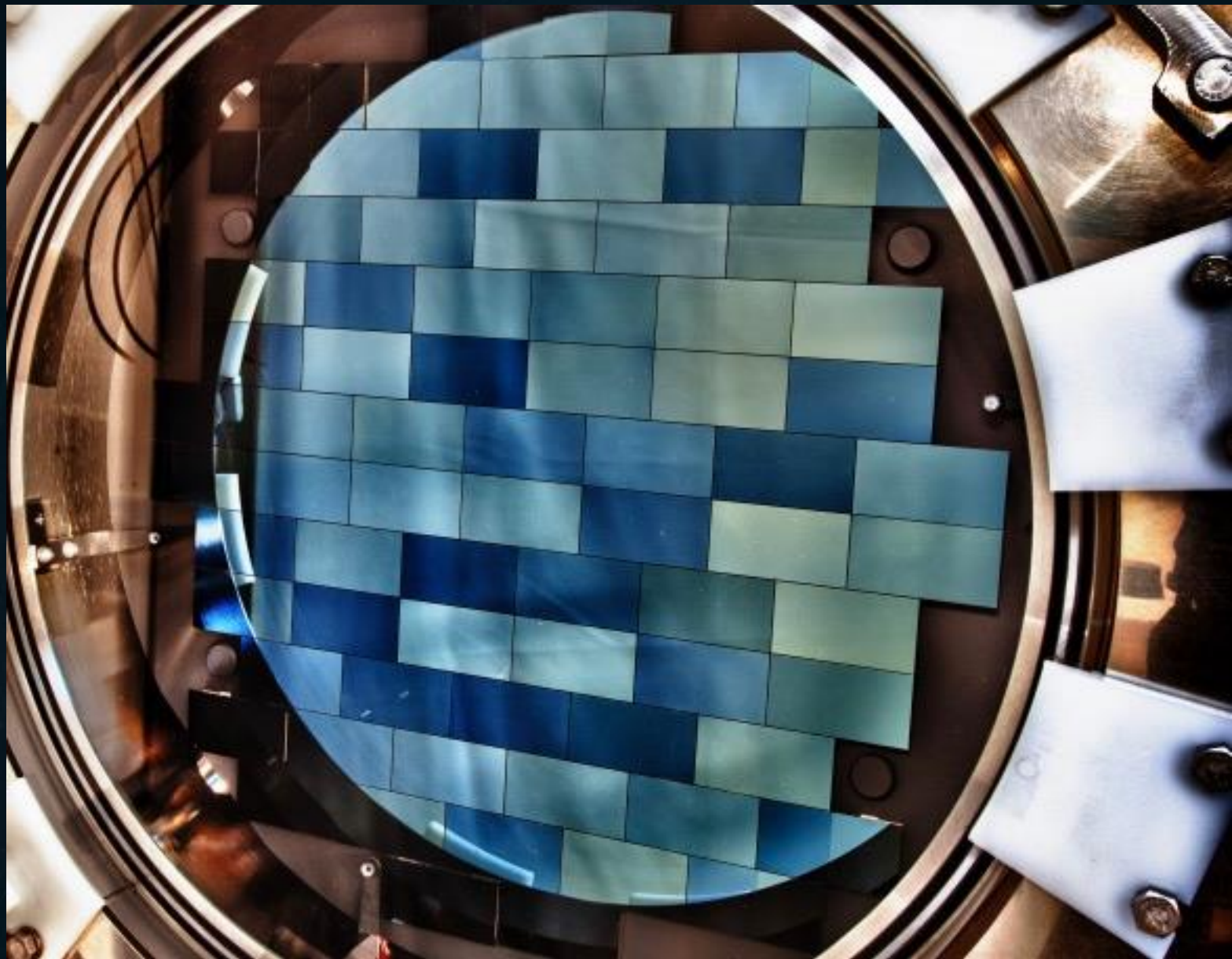
BARYON ACOUSTIC OSCILLATIONS



2

Measure ruler size





THE DARK ENERGY SURVEY (DES)

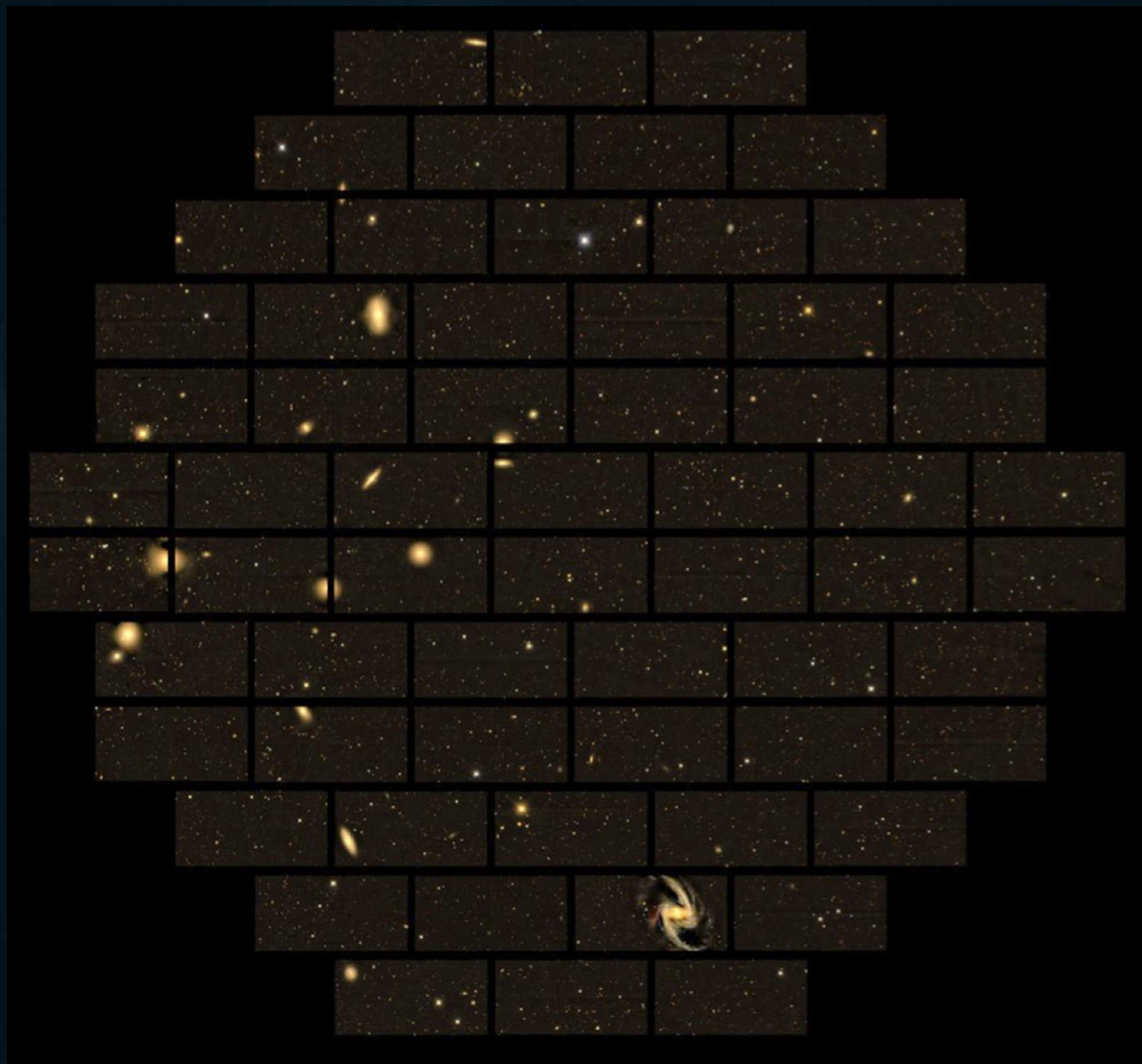
01

Images from Blanco with DECam

543 million galaxies
145 million stars
700,000 asteroids
20,000 supernovae

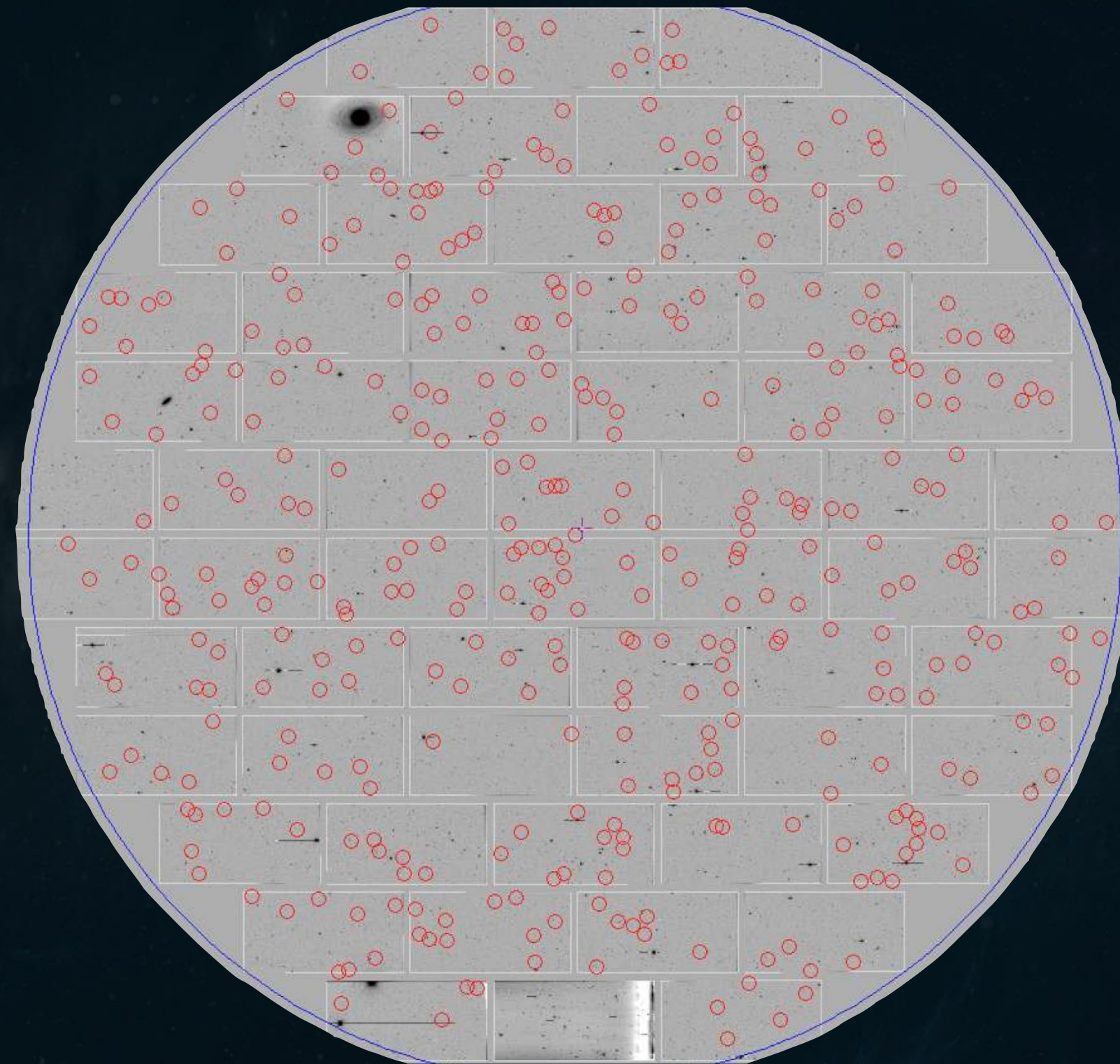
02

Spectra from AAT with 2dF



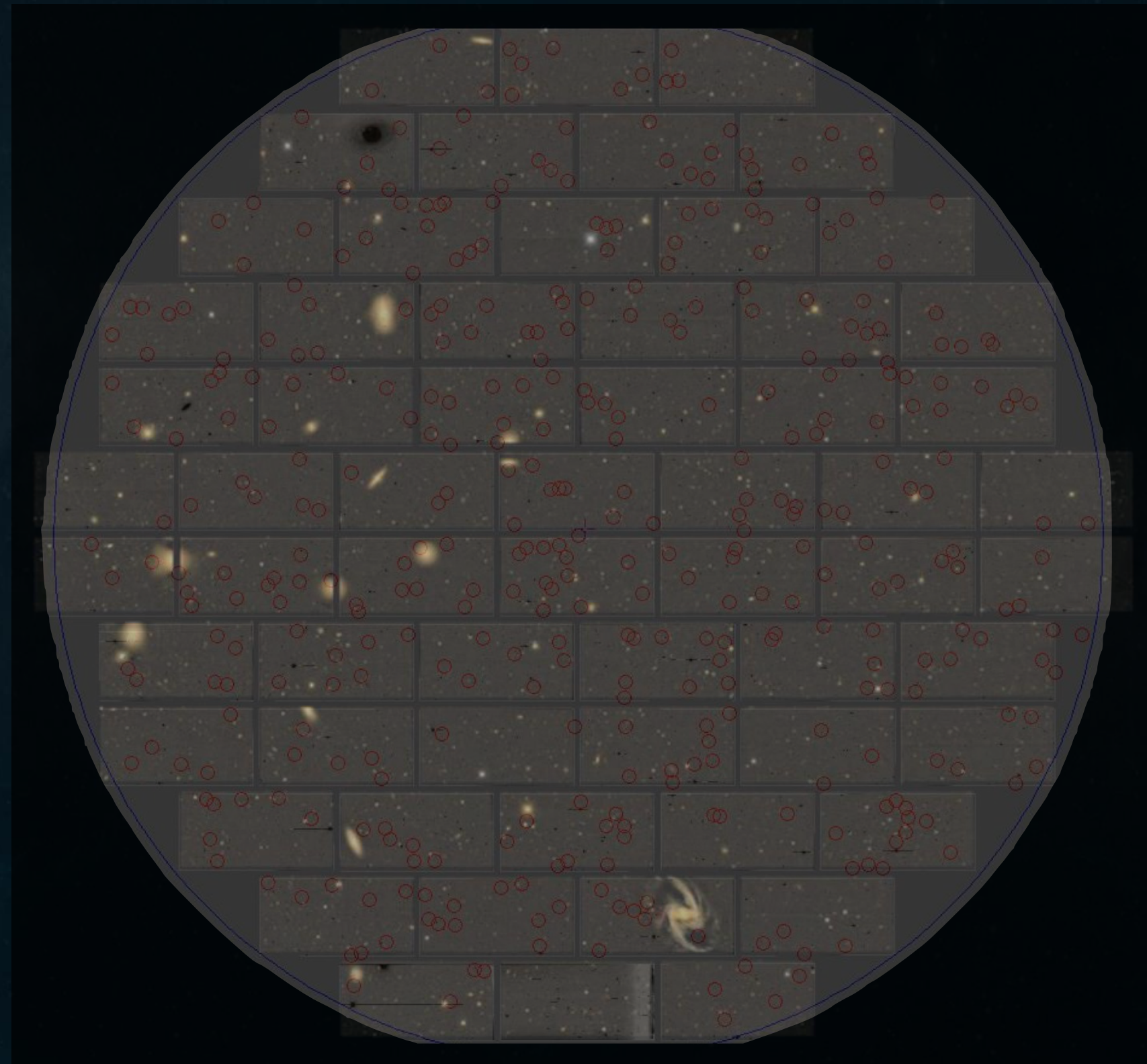
01

Images from Blanco with DECam

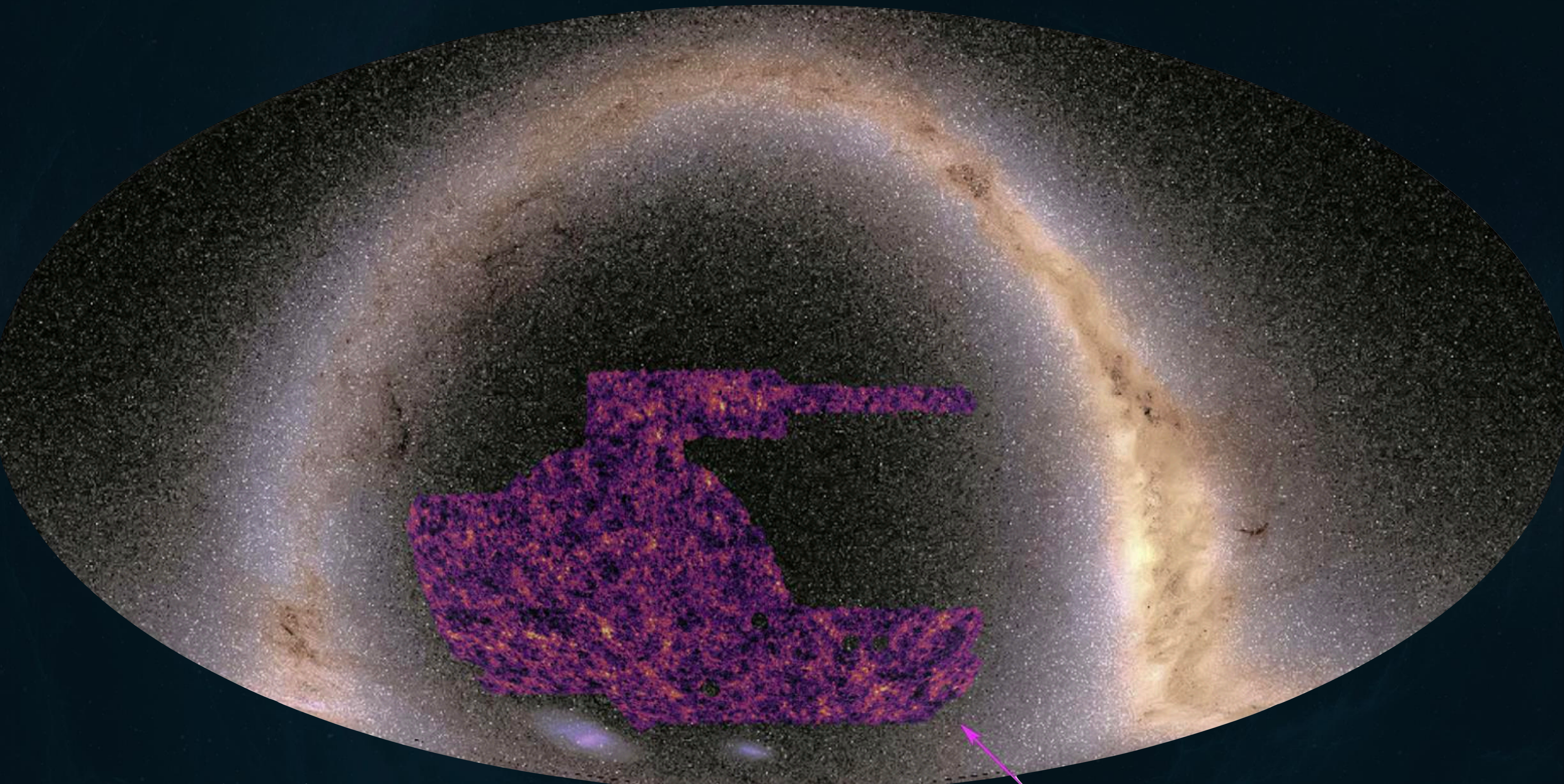


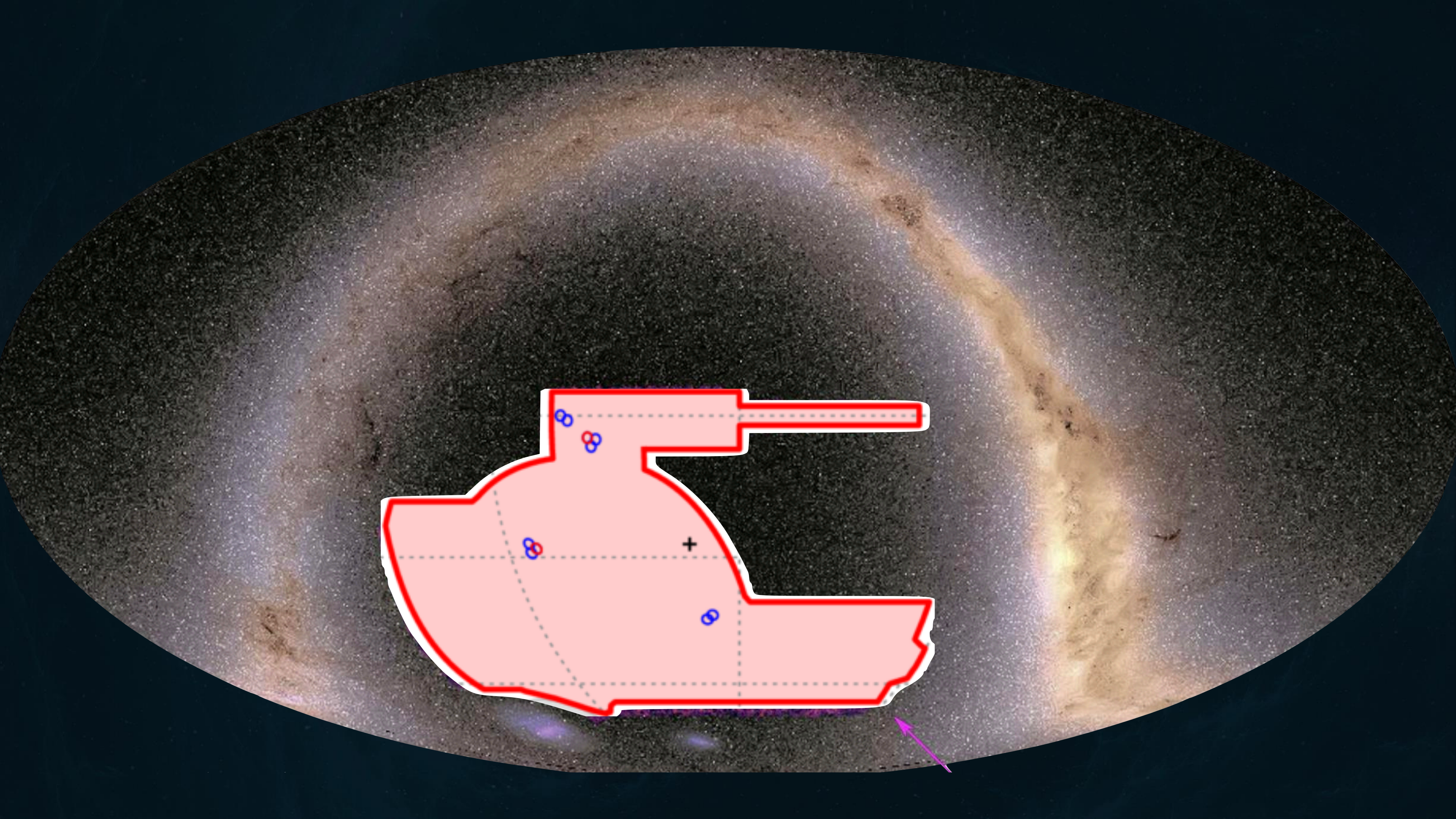
02

Spectra from AAT with 2dF



Identical Fields of View

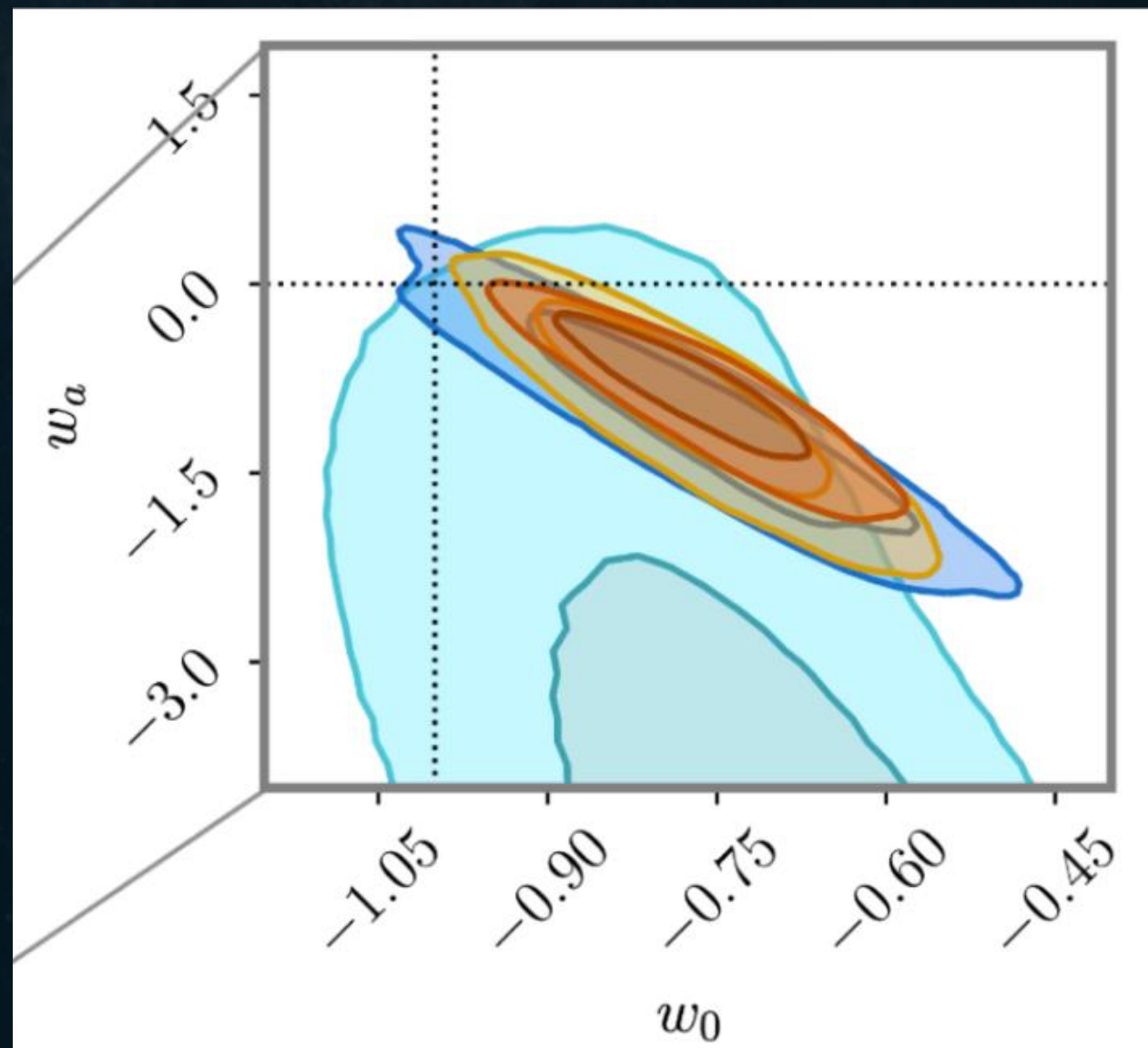






DES RESULTS

The **largest** and **deepest** SN sample from a **single telescope** ever compiled



The data

Approx 1500 new high-redshift supernovae

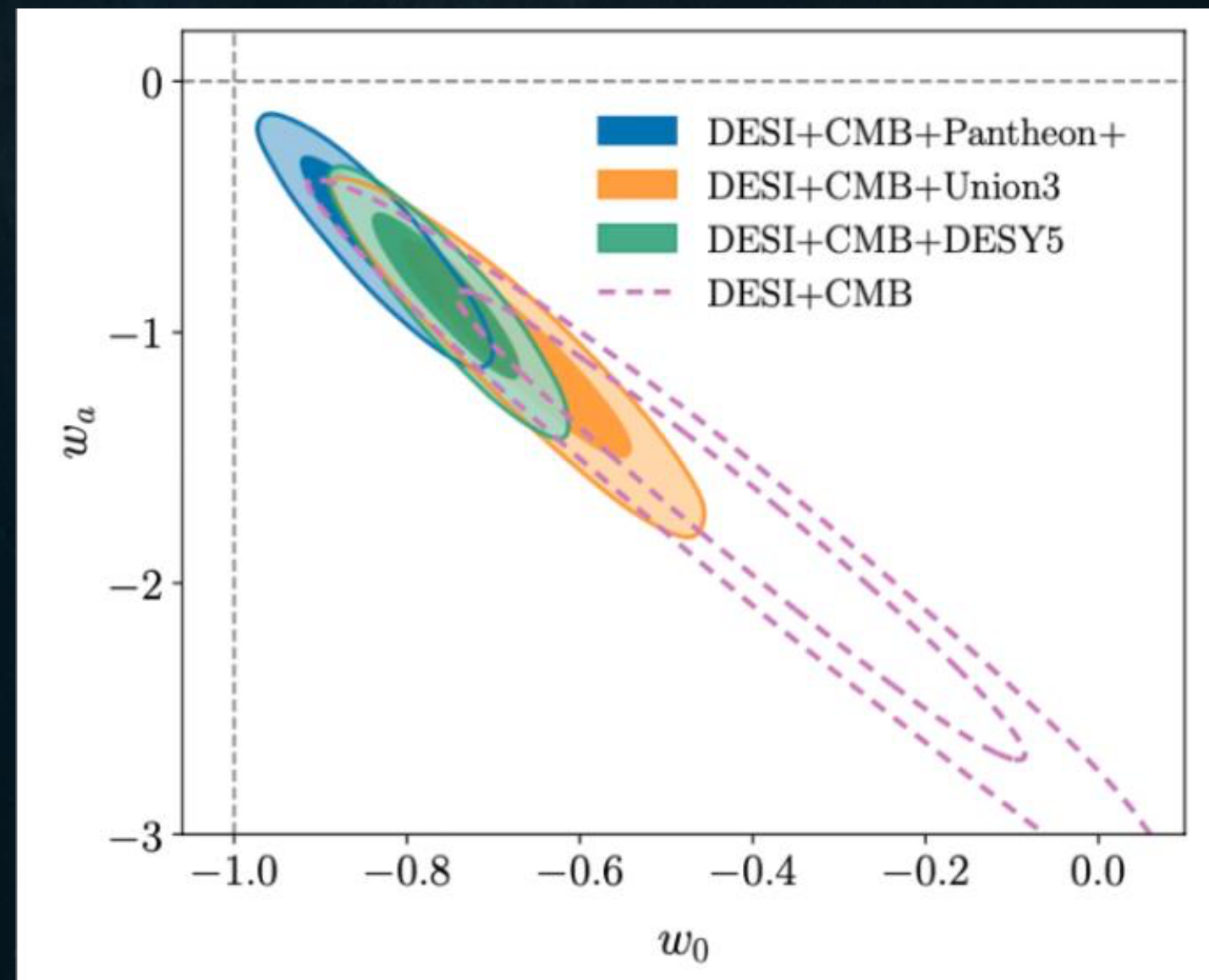
80% of SNe Ia ($z > 0.5$)

The result

Found hints that **dark energy may change with time**

DESI RESULTS

The **largest** and **deepest** BAO sample ever compiled



The data

Approx 6 million galaxies in DR1

Approx 14 million galaxies in DR2

The result

Found more hints that **dark energy may change with time**



01. Confusion

Confusion around which supernova data set to use.

02. Low- z

DES didn't include much of the low- z data, which meant studies that needed many low- z supernovae couldn't use that data set.

03. Upgrades

Methods have improved since Pantheon+ was published, so it needed to be updated.

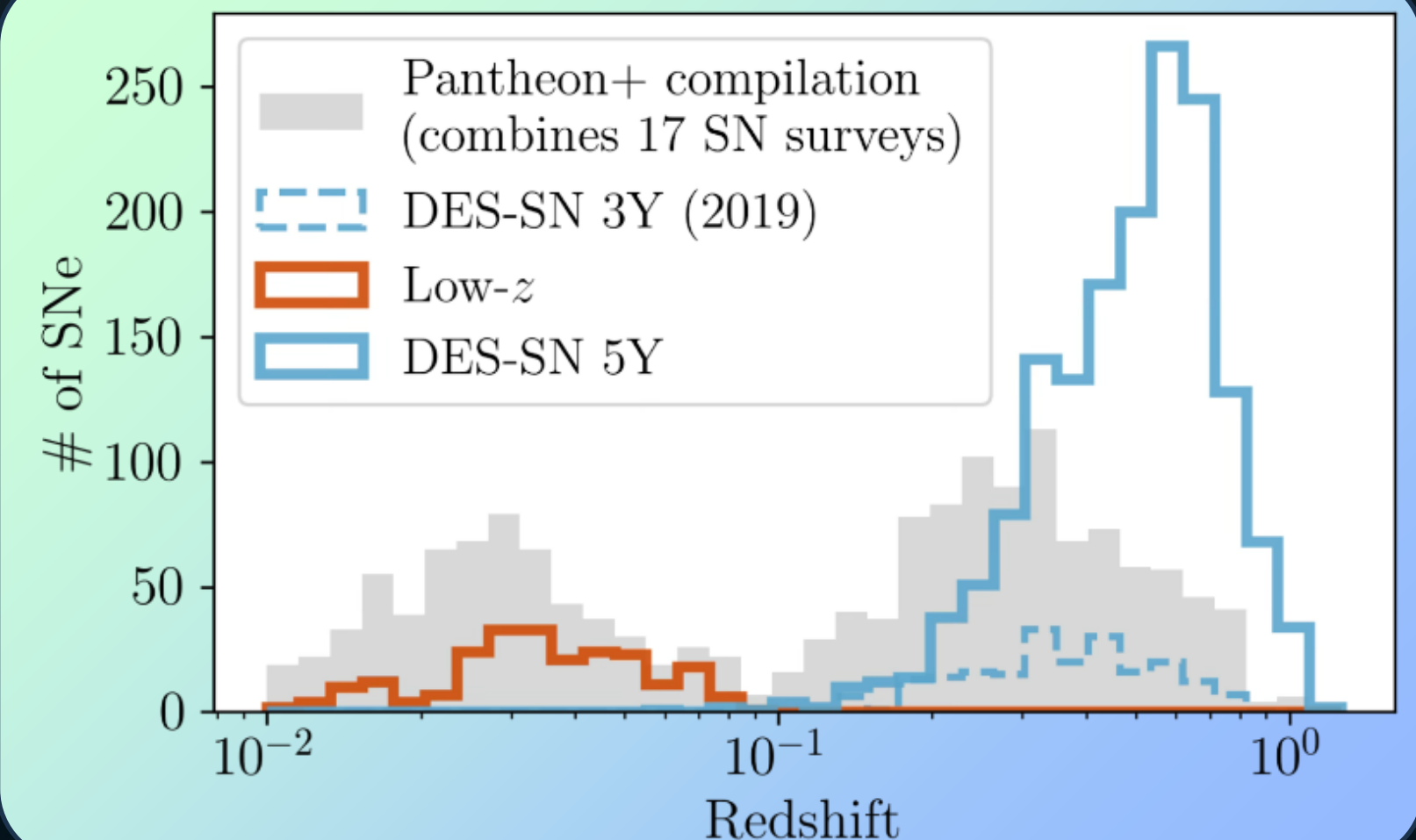
04. BBC+Unity

Wanted to apply both analysis methods, BBC and Unity.

05. Statistics

We can get greater statistical power with more supernovae consistently analysed.

MOTIVATION FOR UNITE



Ryan Camilleri

The University of Queensland

On behalf of the Unite team!

UNITE ANALYSIS

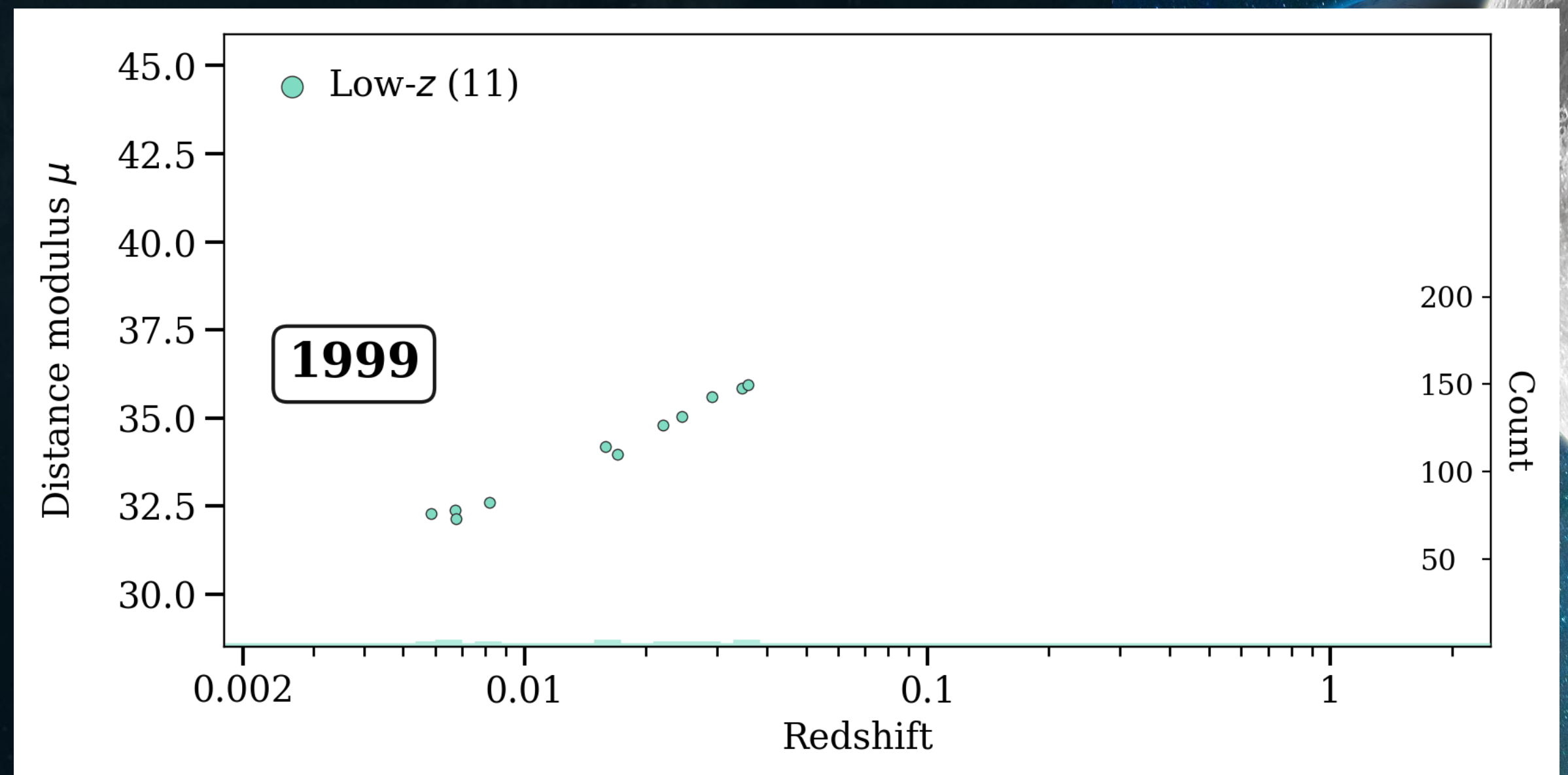
Camilleri et al: 2609.05053
Lee et al.: 2609.05321

Data URL coming soon!



THE UNITE SAMPLE

Combines Pantheon+ and DES-SN5YR
= 18 Surveys, consistently analysed



HUBBLE DIAGRAM

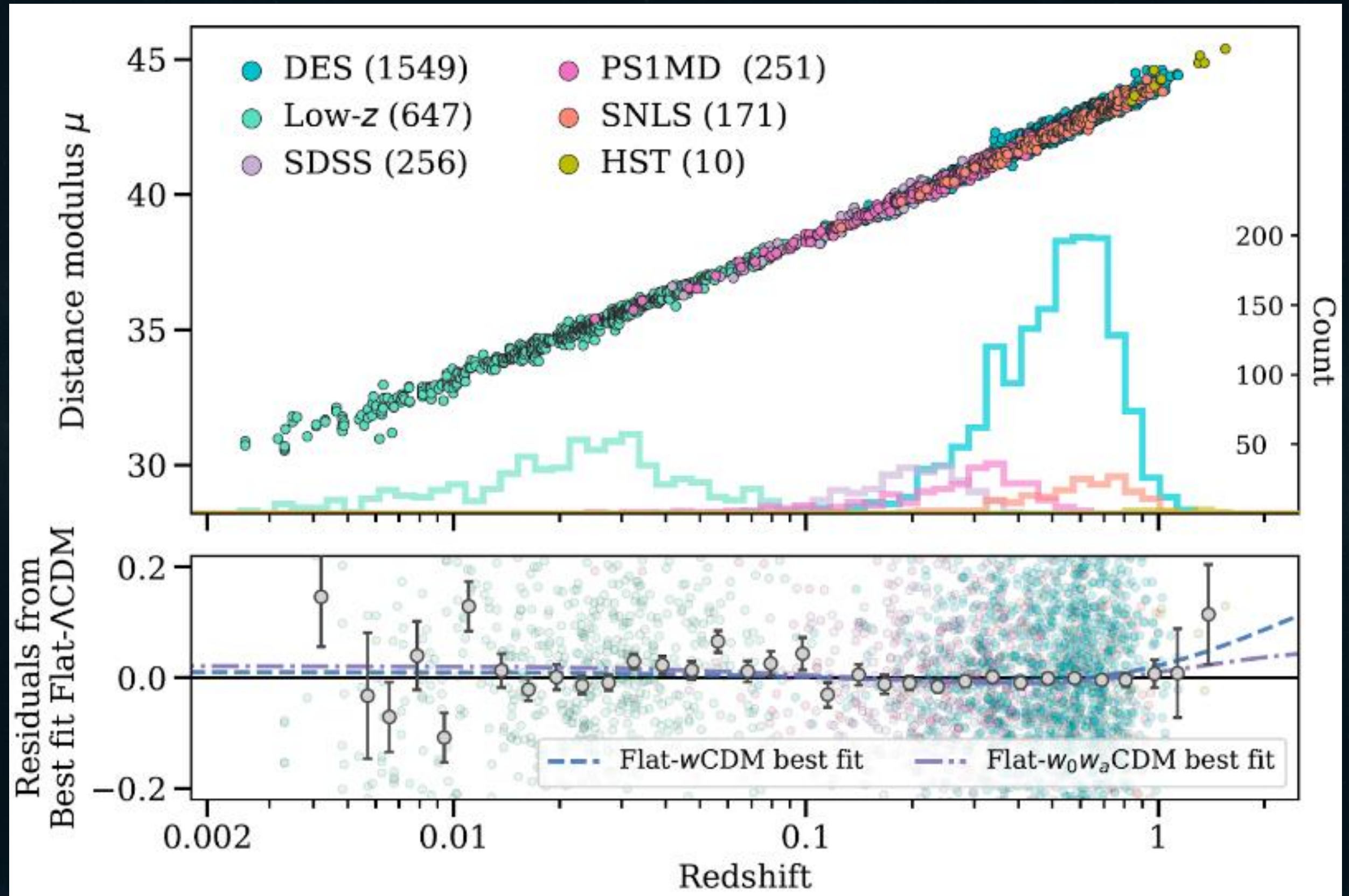
2884 Type Ia Supernovae

Dovekie Calibrations

New Host Masses

Weak-lensing Corrections

Type-Ia Probability > 0.8



HUBBLE DIAGRAM

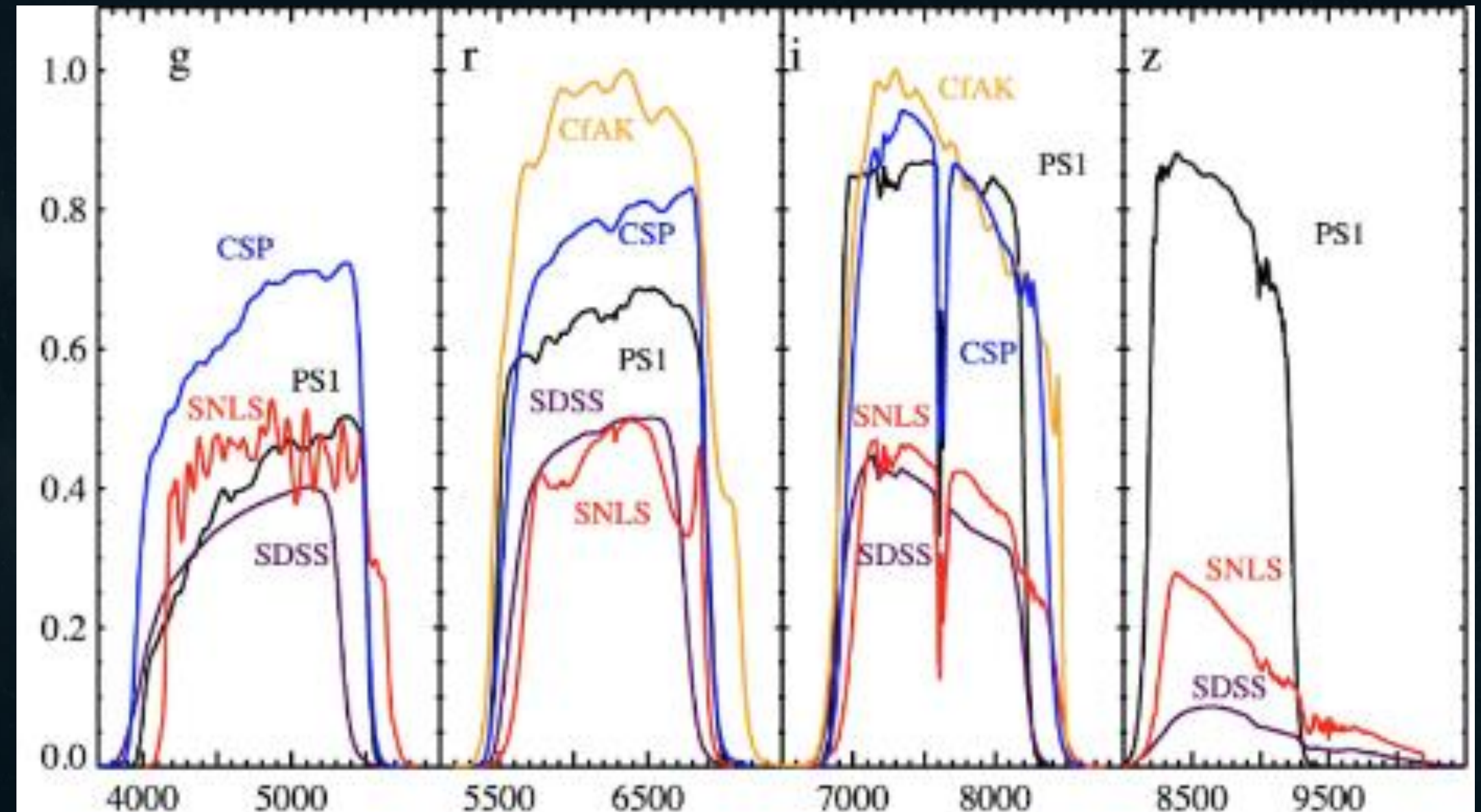
2884 Type Ia Supernovae

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New Host Masses

Weak-lensing Corrections

Type-Ia Probability > 0.8



Dovekie cross-calibration by Popovic et al. 2025

HUBBLE DIAGRAM

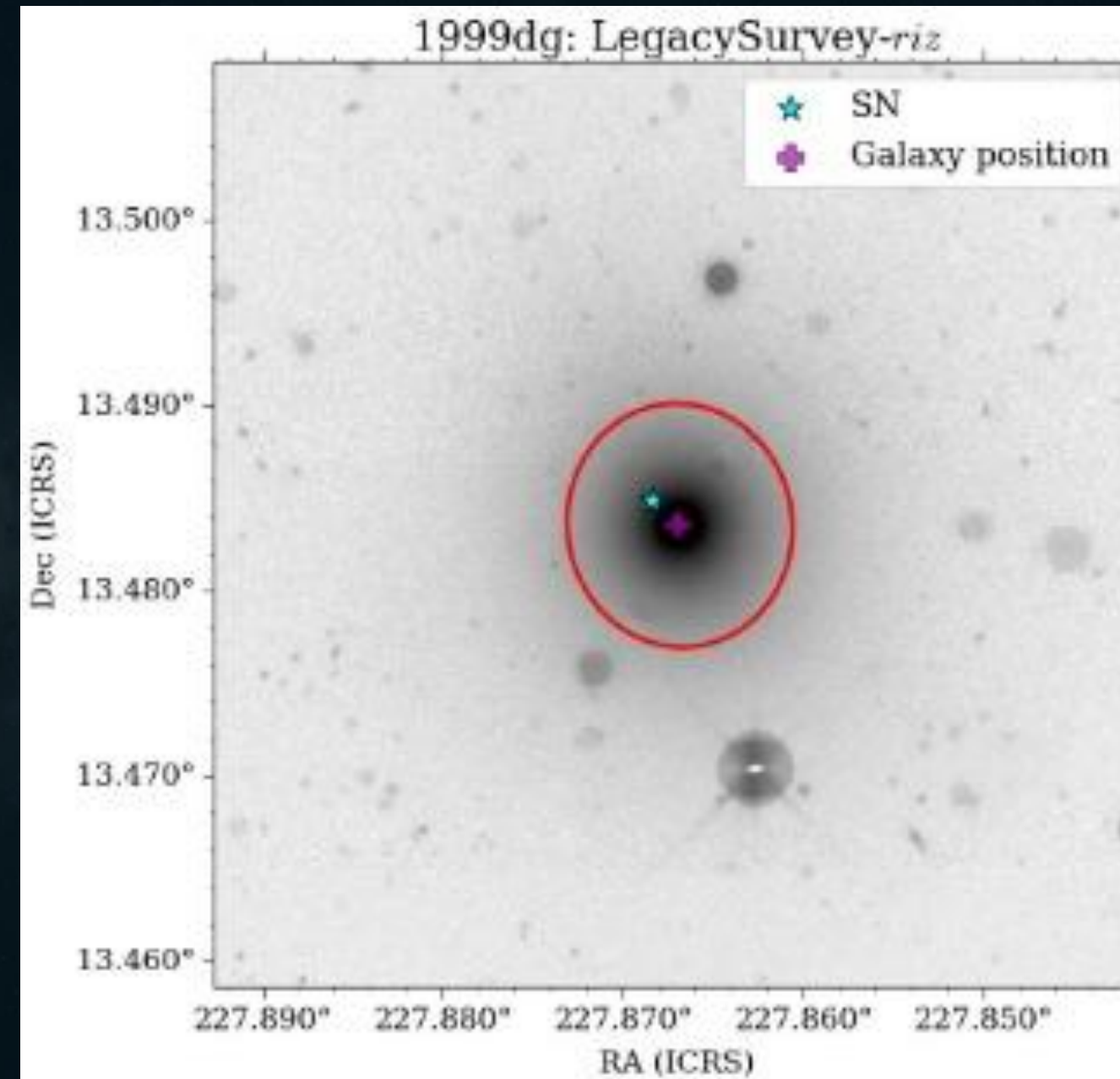
2884 Type Ia Supernovae

Dovekie Calibrations

New Host Masses

Weak-lensing Corrections

Type-Ia Probability > 0.8



Consistent host-galaxy masses by Lee et al. 2026

HUBBLE DIAGRAM

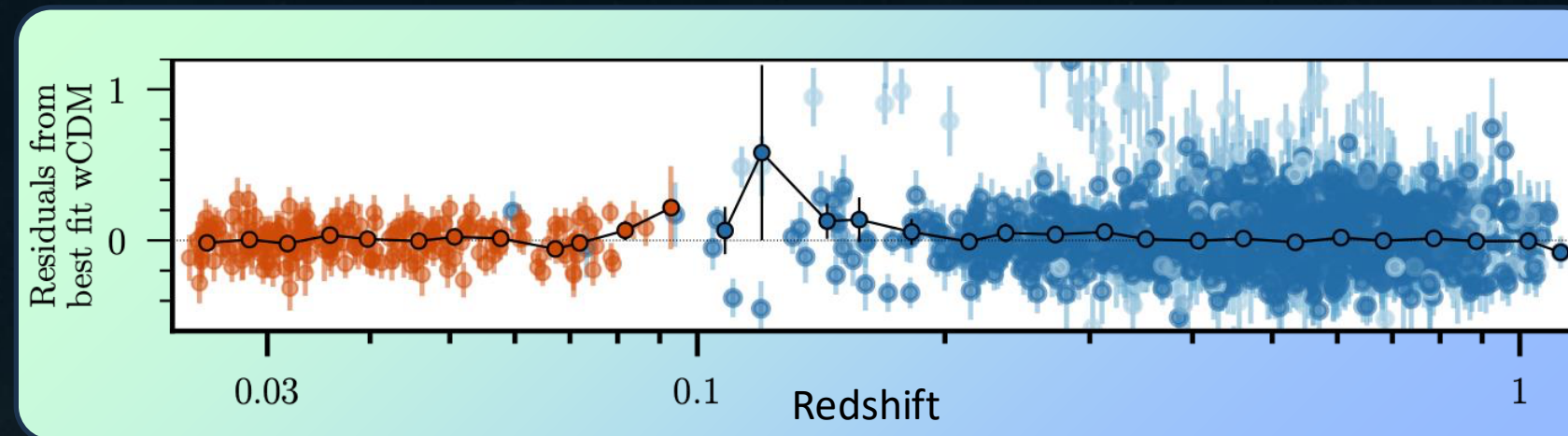
2884 Type Ia Supernovae

Dovekie Calibrations

New Host Masses

Weak-lensing Corrections

Type-Ia Probability > 0.8



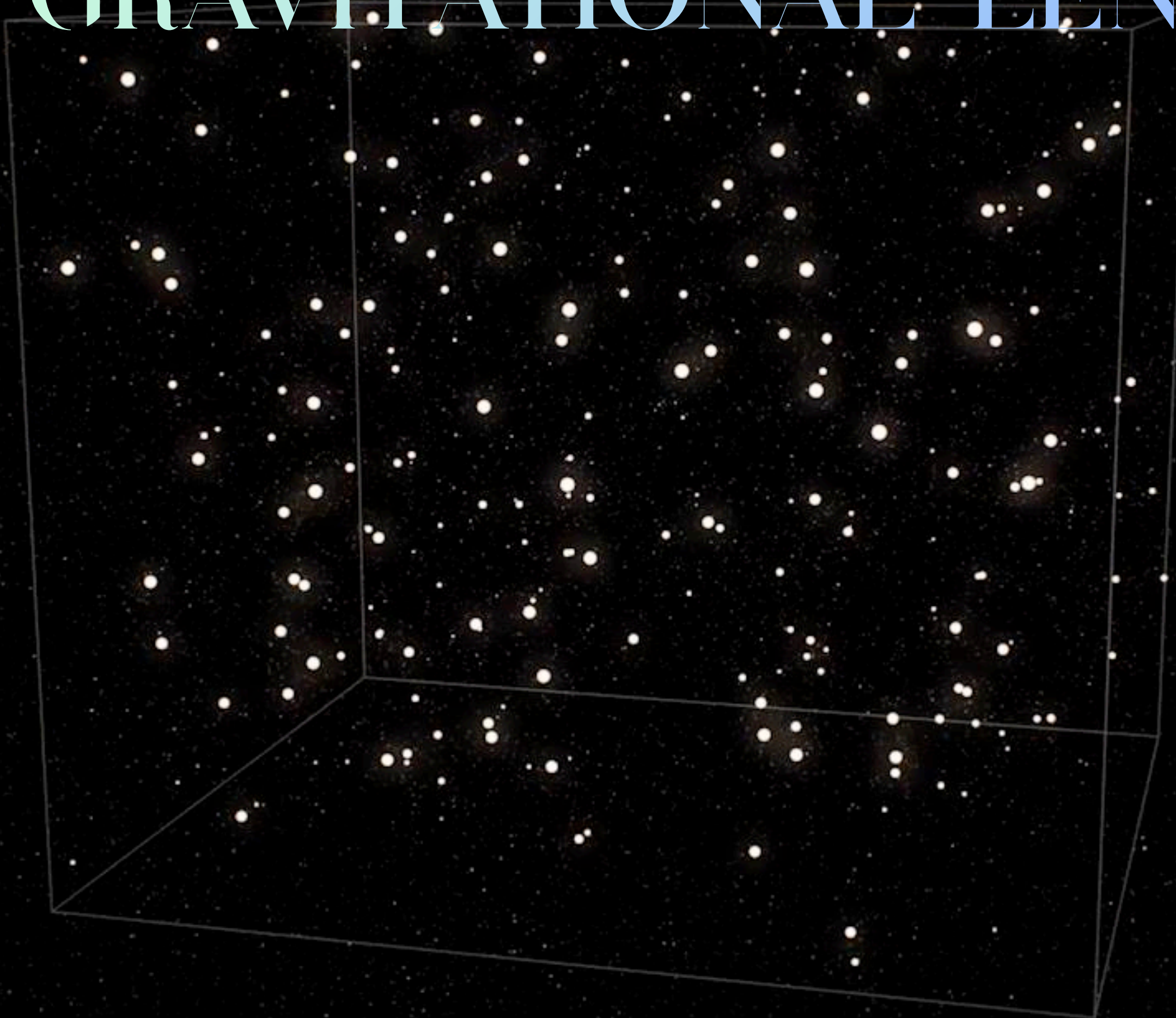
Updated lensing treatment by Shah et al. 2024

STRONG GRAVITATIONAL LENSING



WEAK GRAVITATIONAL LENSING

Galaxies



HUBBLE DIAGRAM

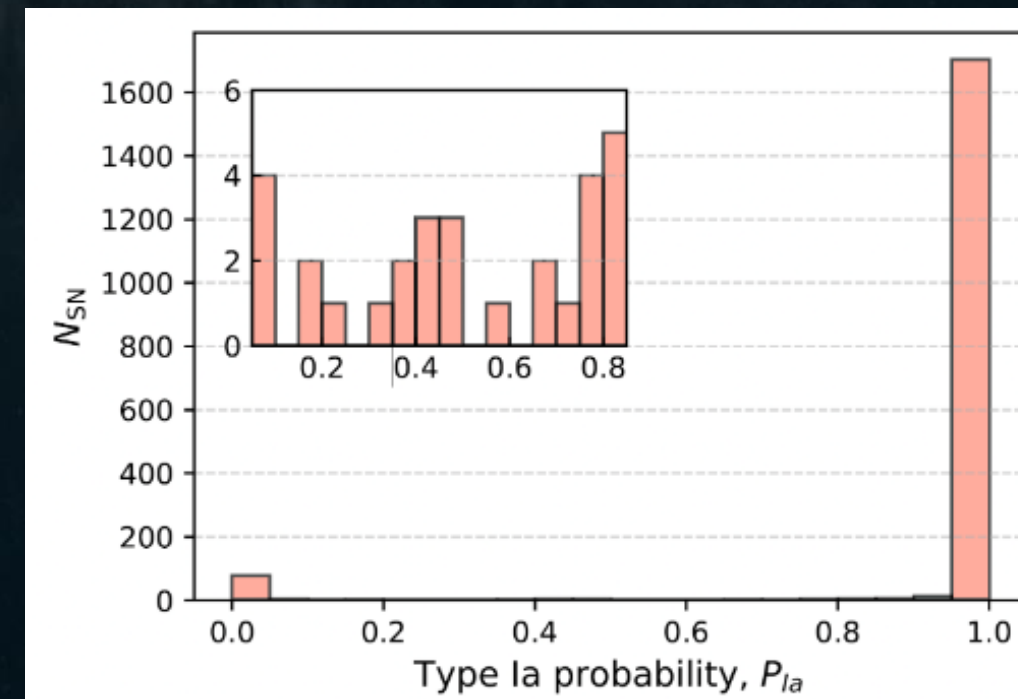
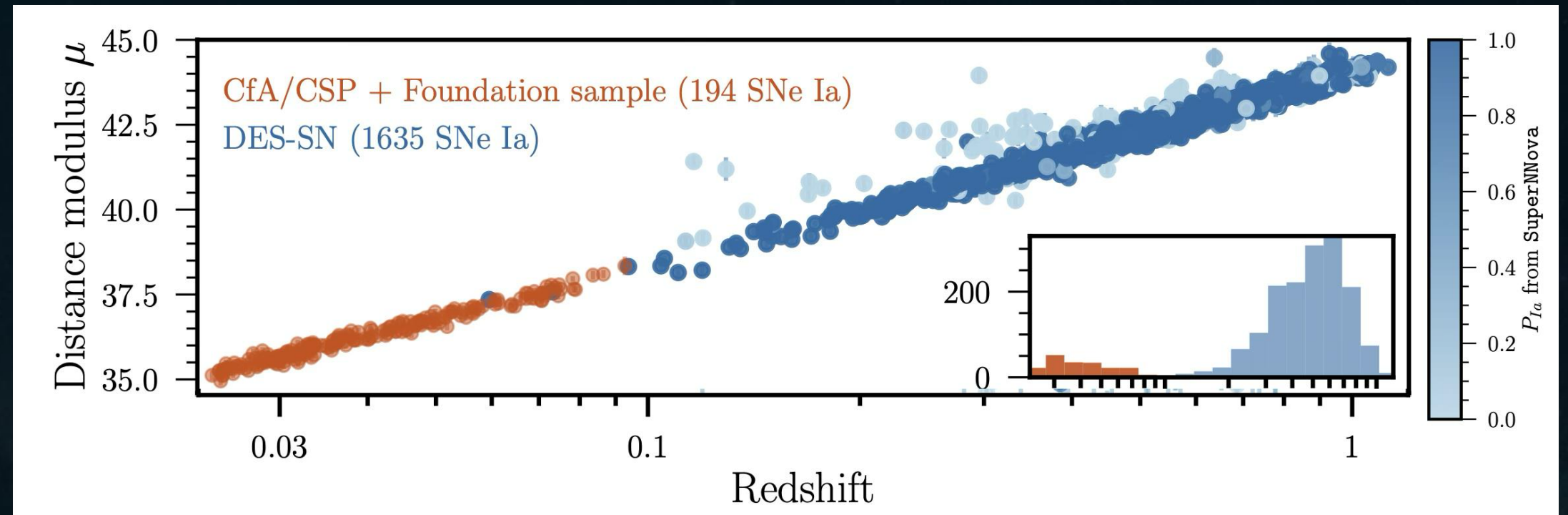
2884 Type Ia Supernovae

Dovekie Calibrations

New Host Masses

Weak-lensing Corrections

Type-Ia Probability > 0.8



Added a type Ia Probability Cut

HUBBLE DIAGRAM

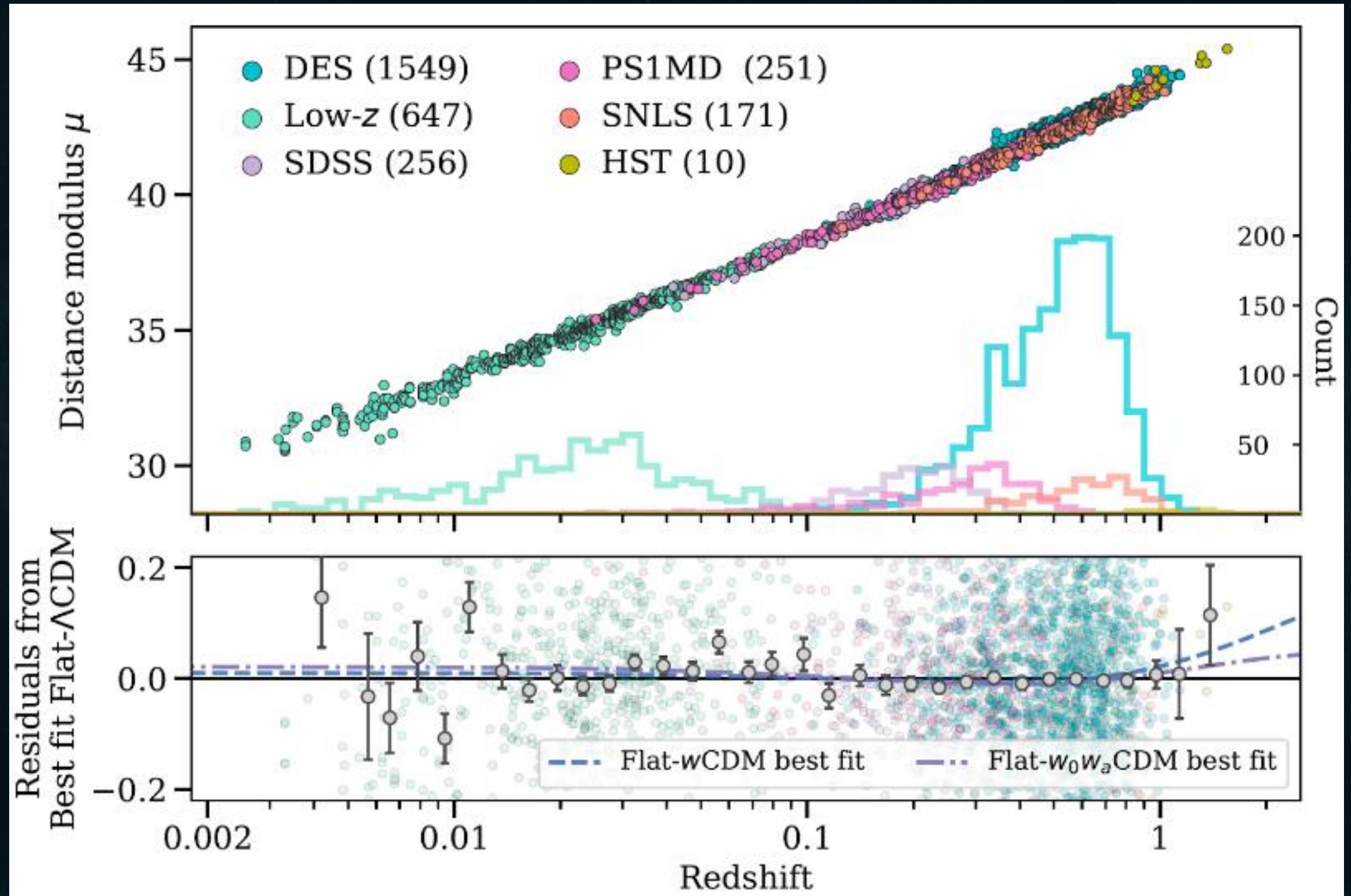
2884 Type Ia Supernovae

Dovekie Calibrations

New Host Masses

Weak-lensing Corrections

Type-Ia Probability > 0.8



UPGRADES GALORE!

Consistently combines the two largest supernova compilations, *bringing all the latest updates to Pantheon+*, including new **host galaxy masses**

01 Light curve fitting

Update Pantheon+ light curve fitting from SALT2 to SALT3

02 Dust model

Update Pantheon+ with latest dust model

03 Dovekie

Update Pantheon+ cross-calibration to use Dovekie

04 Systematics

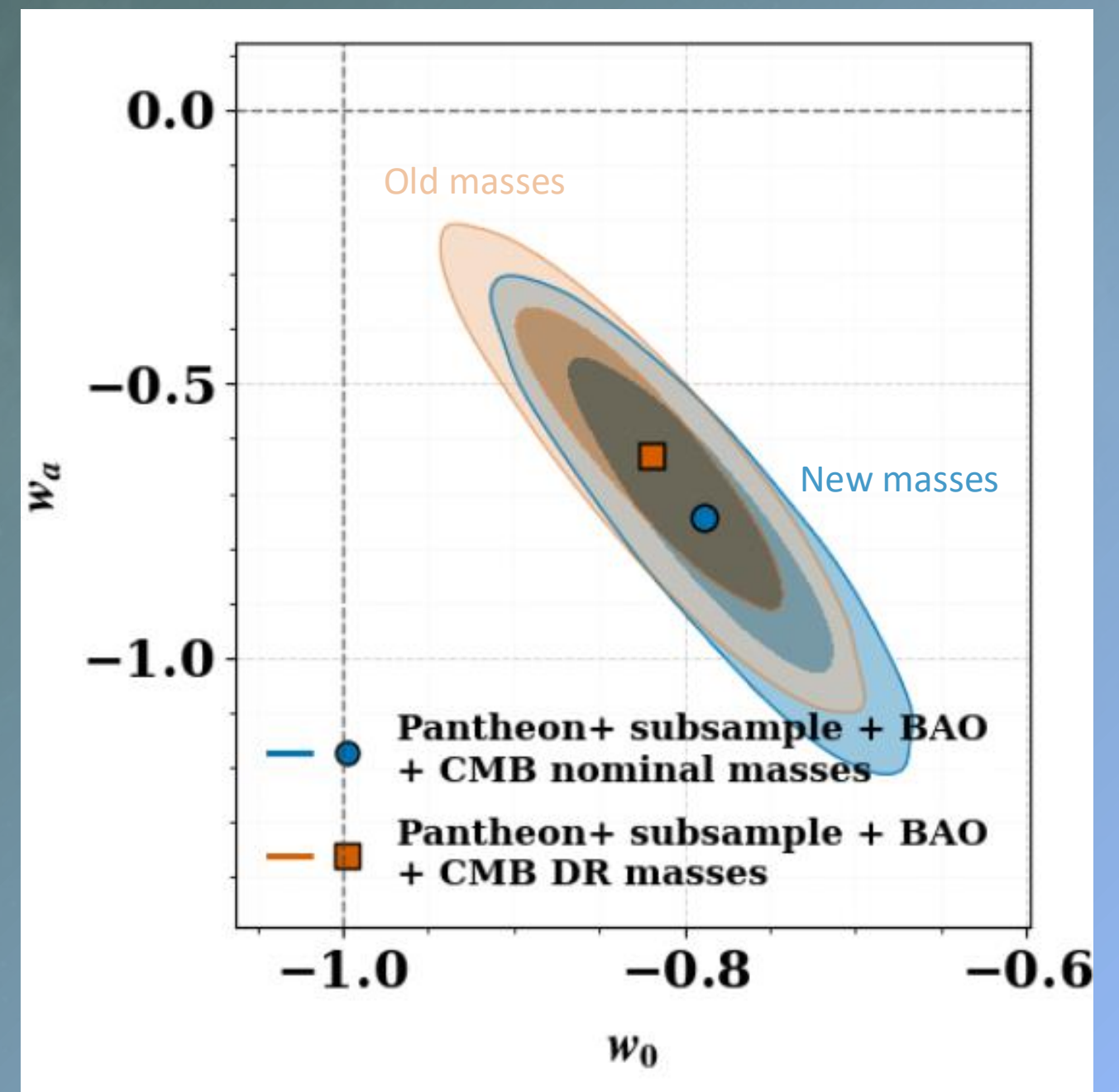
Consider additional systematics such as host mass uncertainties

05 Add lensing

Add gravitational lensing correction

06 Host masses

Derive new host galaxy masses for every host in Pantheon+ and DES



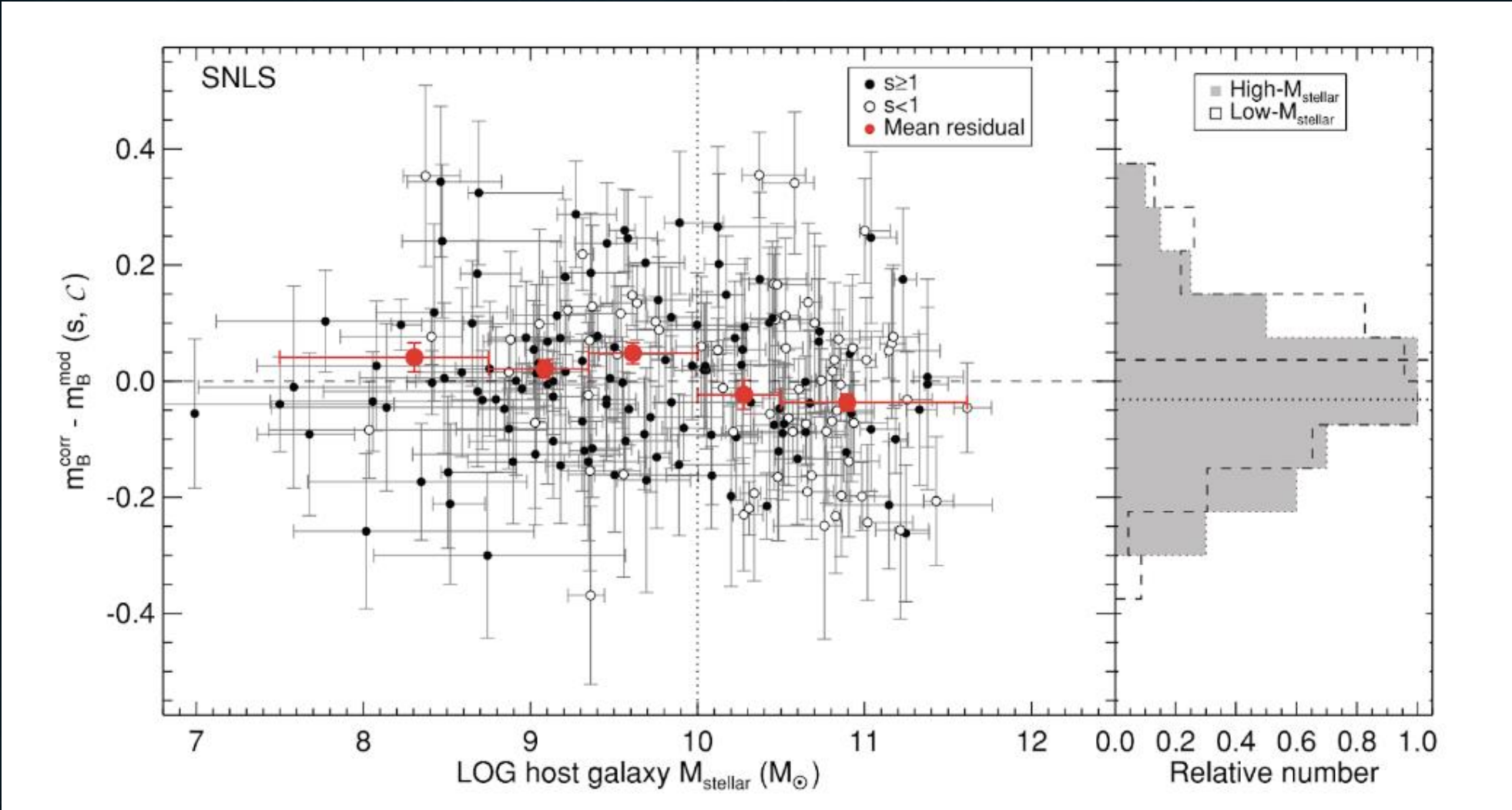
Jason (Jaemyoung) Lee

On behalf of the Unite team!

HOST MASS

Lee et al.: 2609.05321

Data URL coming soon!



THE MASS STEP

SNe Ia in **more massive galaxies** are **brighter by 0.05 to 0.10 mag** after standardization

$\gamma = 0.040 \pm 0.006$

$G_{host} = \pm \frac{1}{2}$
 + for high-mass hosts
 - for low-mass hosts

apparent magnitude

$$\mu_B = m_B^* - M_B + \alpha x_1 - \beta C + \gamma G_{host} - \Delta\mu_{lens} + \Delta\mu_{bias}$$

absolute magnitude

stretch scale

colour scale

Measured from each supernova light curve

Measured for the population

host mass factor

weak lensing correction

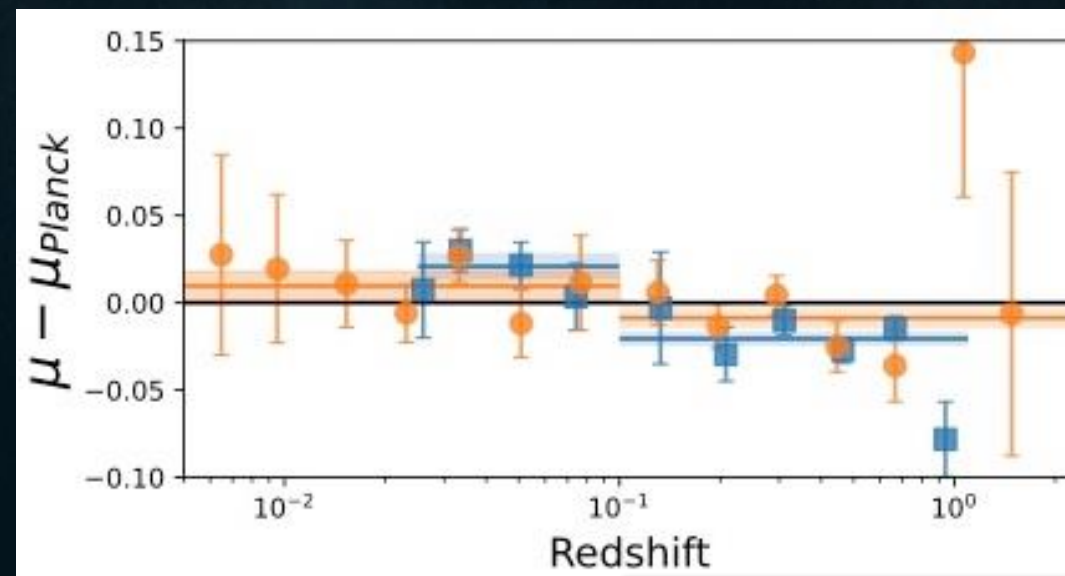
Measured from each host and l.o.s.

WHY WE CARE

It impacts cosmology

Vincenzi et al. 2025

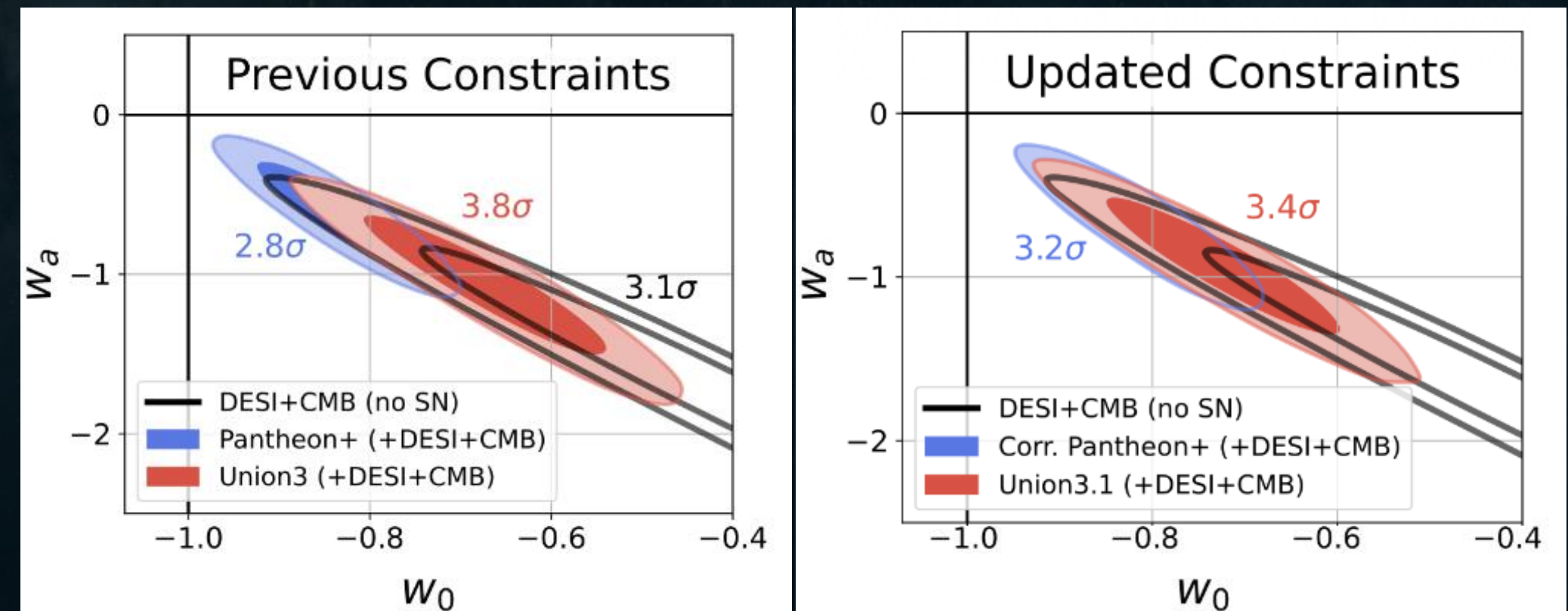
Finds reverting to Pantheon+ host stellar mass estimations gives a 0.010 change to high-z vs low-z distance moduli (larger than expected given uncertainties).



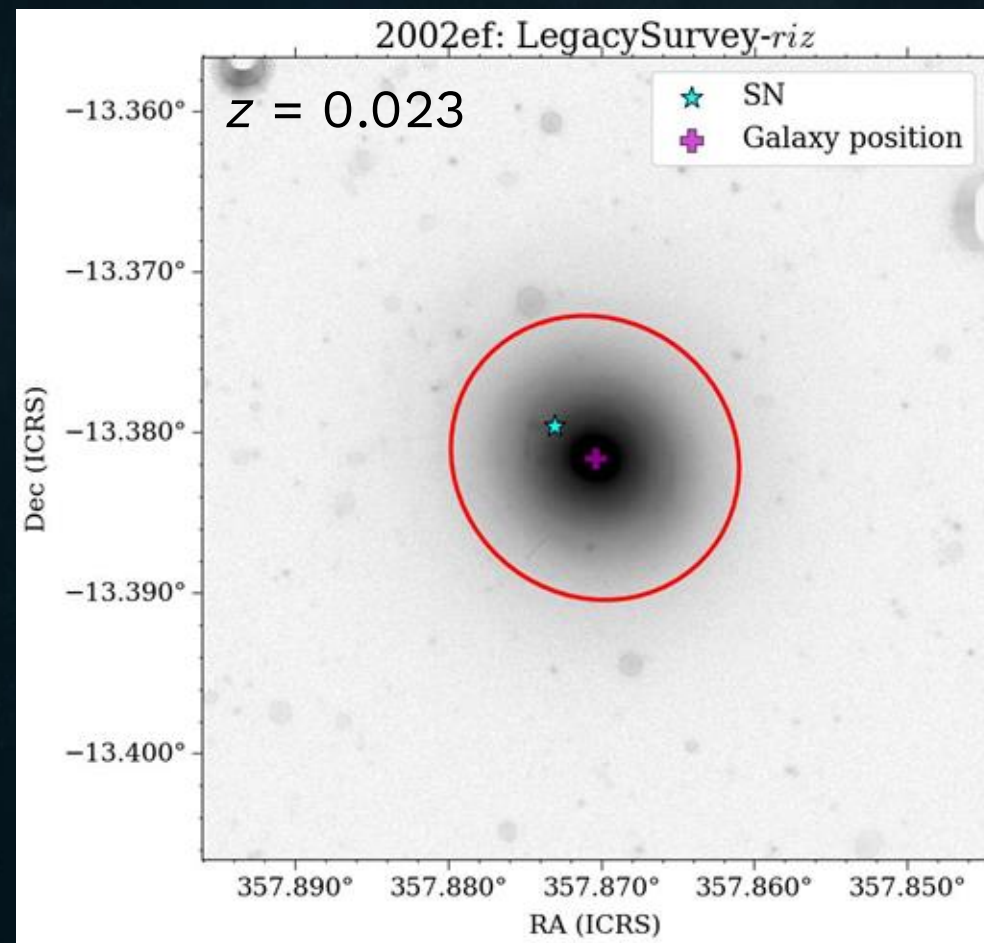
DES-SN5YR
Pantheon+
DES-SN5YR offset from Planck:
Low-z: 0.021 mag
High-z: -0.021 mag
Pantheon+ offset from Planck:
Low-z: 0.010 mag
High-z: -0.009 mag

Hoyt et al. 2026

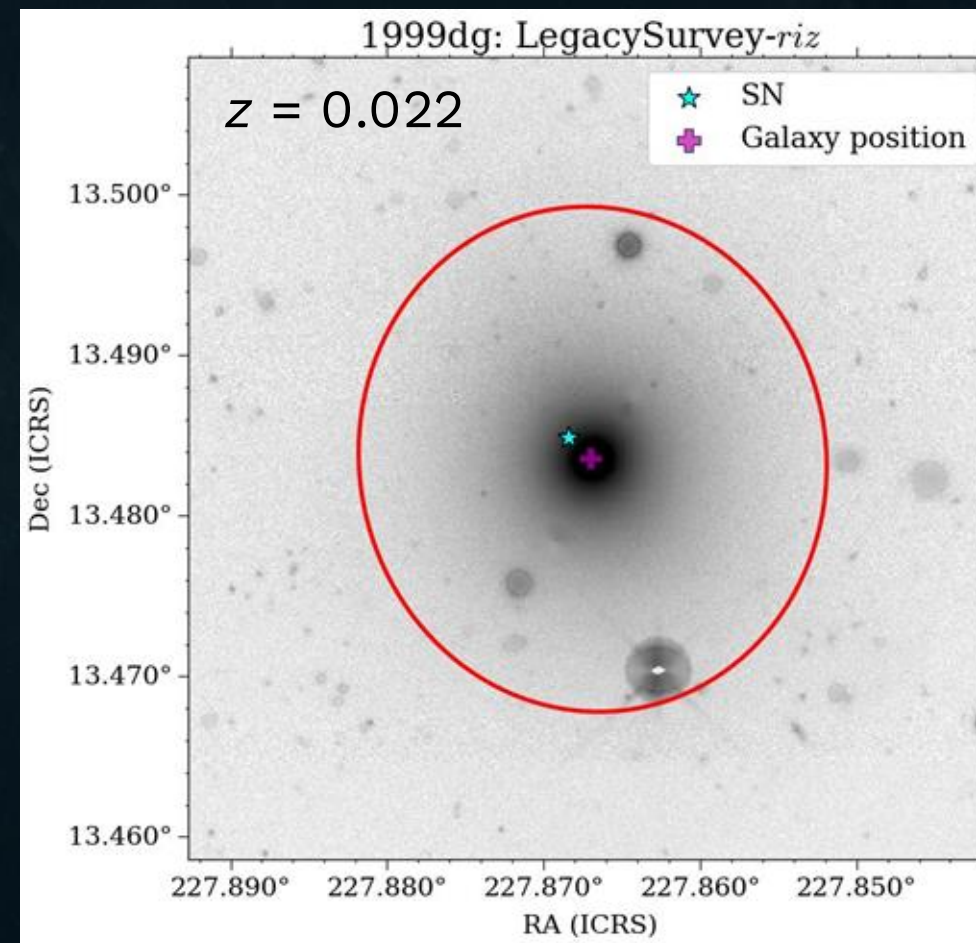
Saw step in masses between low-z and high-z.
Correcting it makes Pantheon+ more consistent with Union3.1



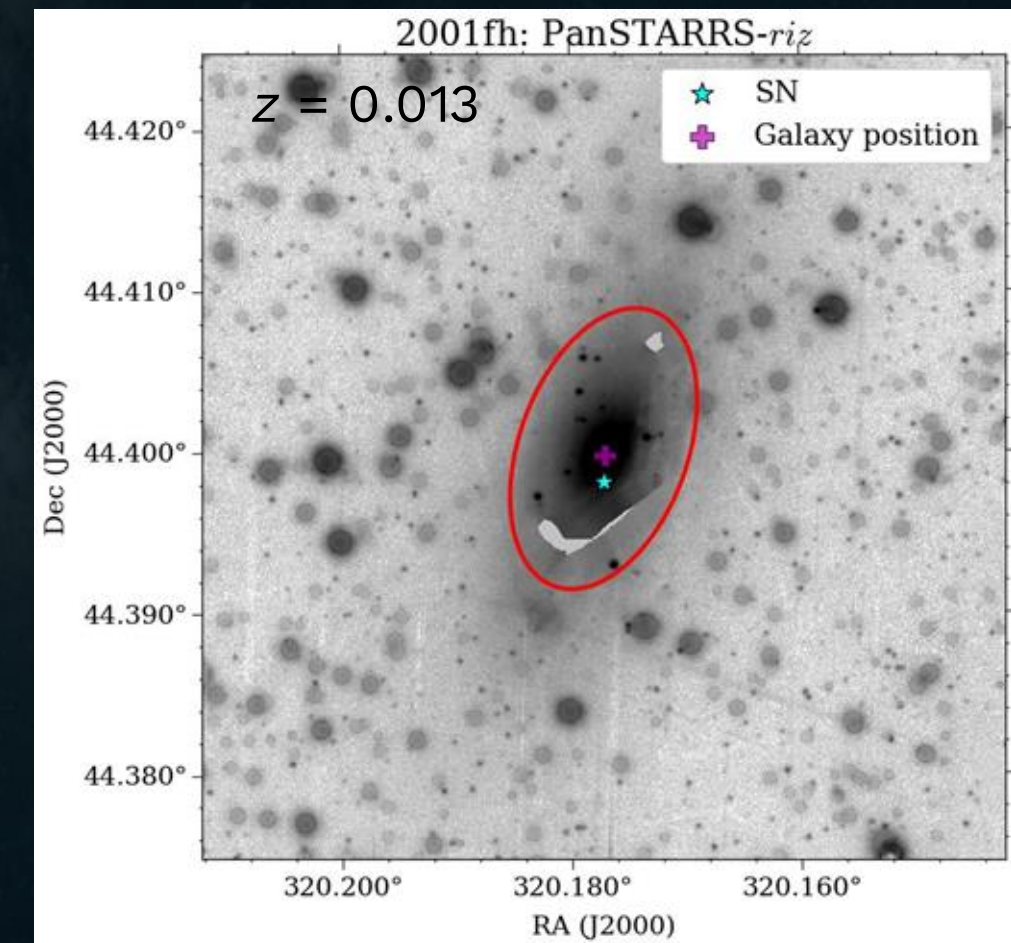
MEASURING HOST- GALAXY STELLAR MASSES



$$M_{\text{stellar}} = 10.69 \pm 0.15$$



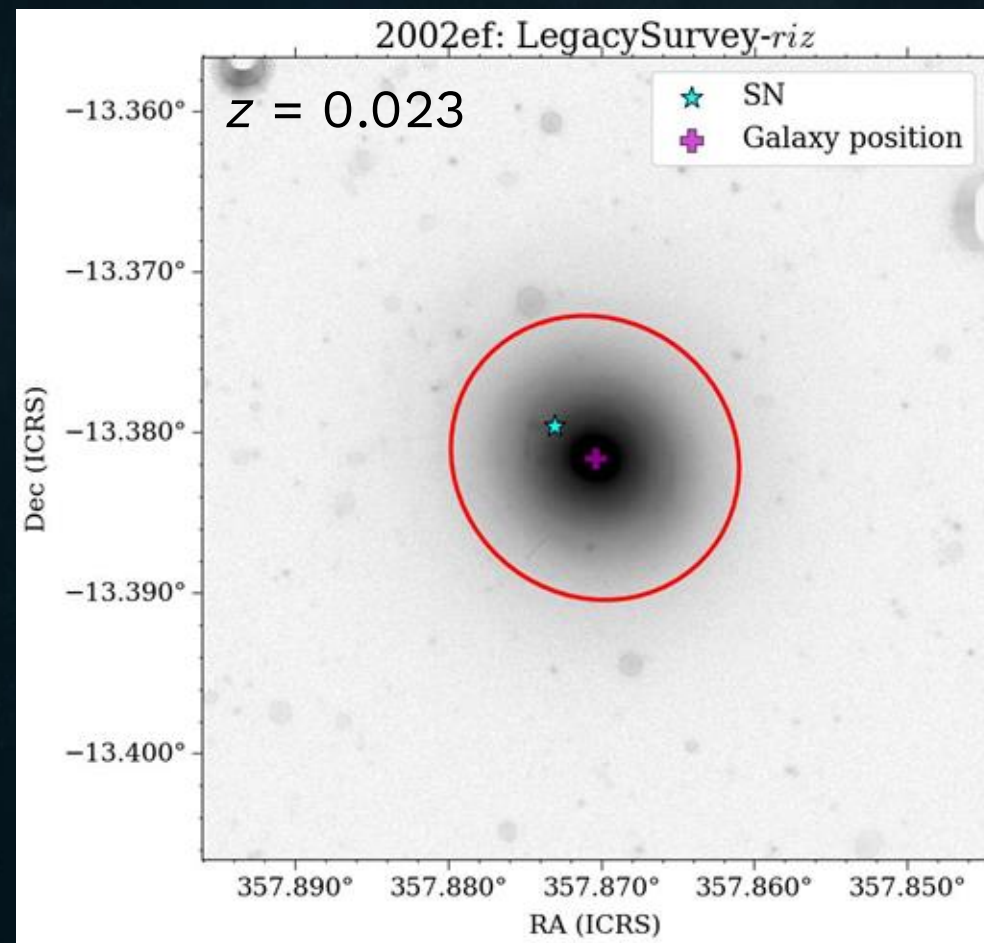
$$M_{\text{stellar}} = 10.67 \pm 0.15$$



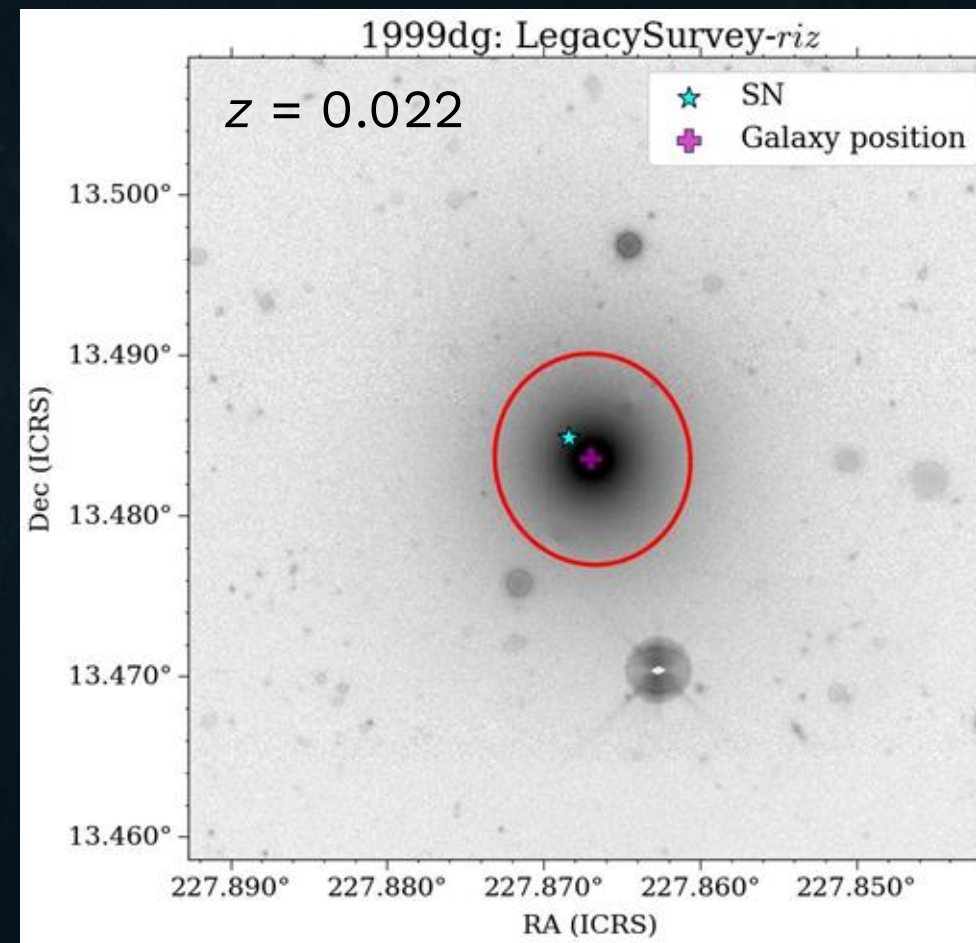
$$M_{\text{stellar}} = 10.65 \pm 0.14$$

All are high-mass galaxies

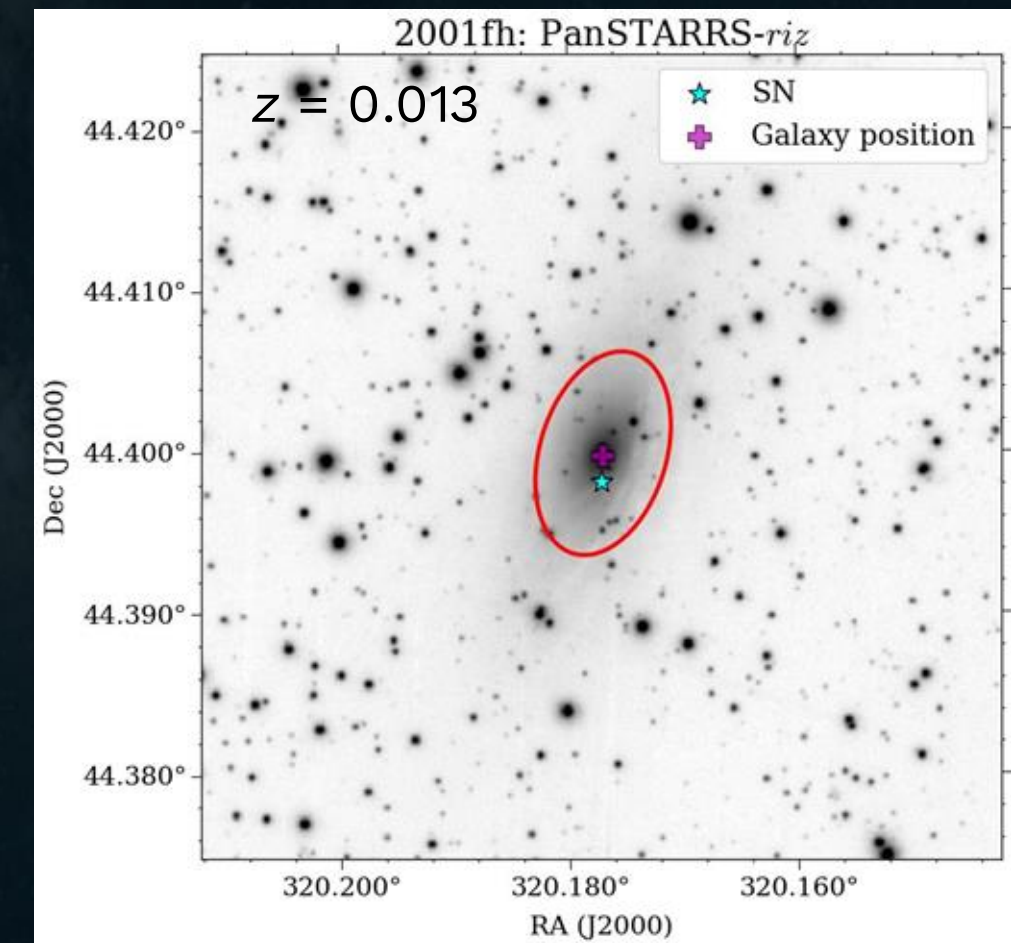
MEASURING HOST- GALAXY STELLAR MASSES



$$M_{\text{stellar}} = 10.69 \pm 0.15$$



$$M_{\text{stellar}} = 10.63 \pm 0.12$$



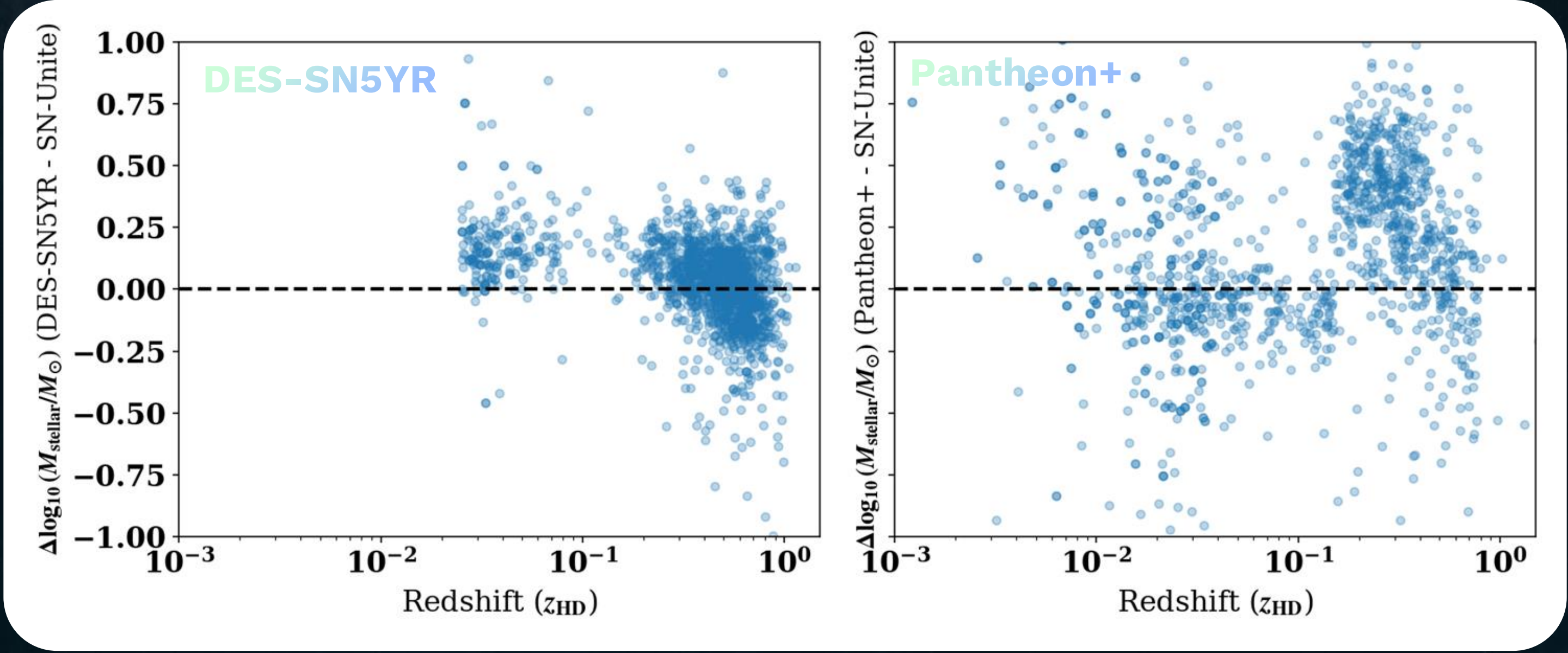
$$M_{\text{stellar}} = 10.53 \pm 0.13$$

Still massive galaxies!

SN-UNITE HOST STELLAR MASSES

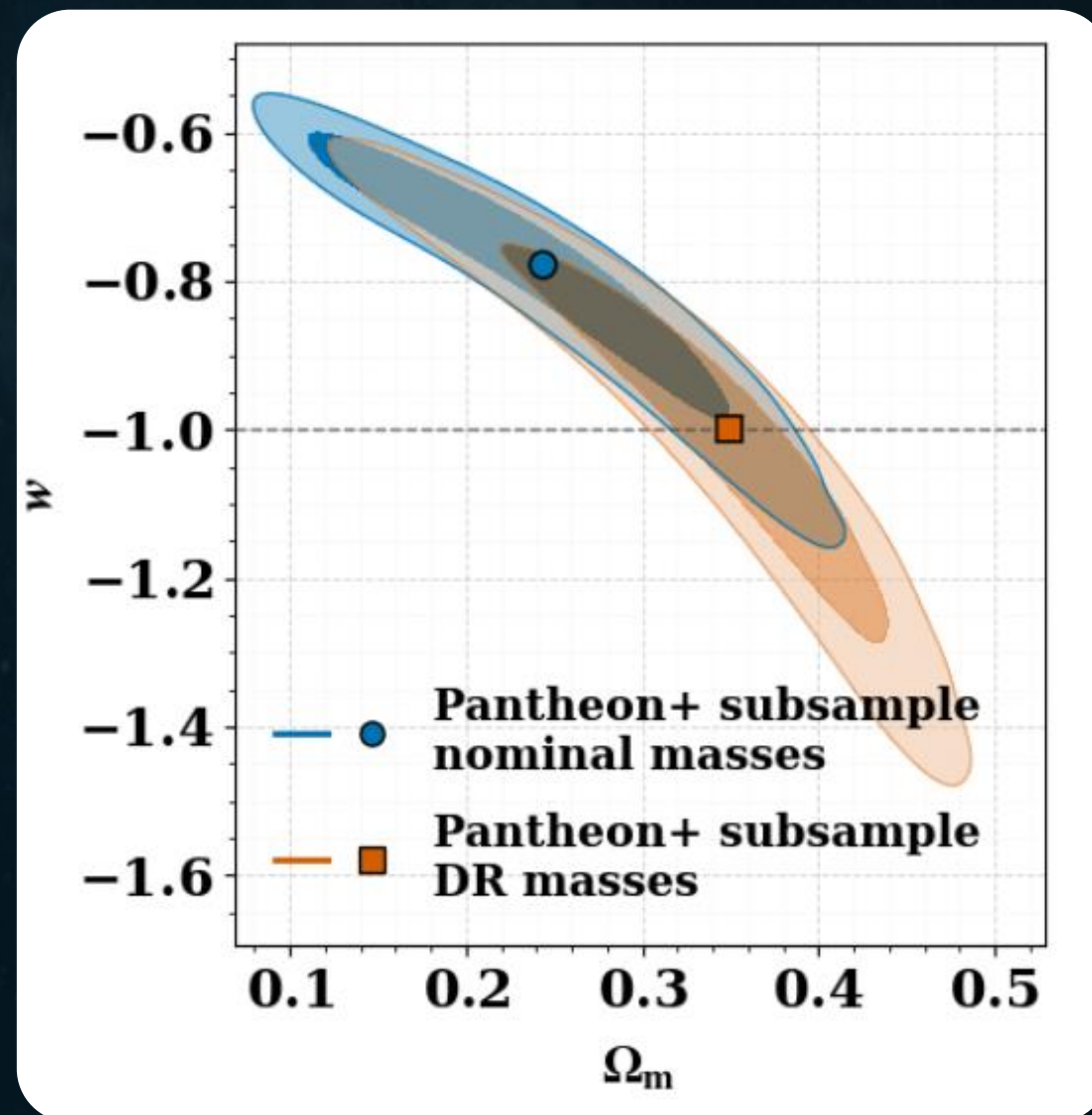
Sample	z bin	High $\rightarrow$ Low	Low $\rightarrow$ High
DES-SN5YR	Low- z ($z_{\text{HD}} < 0.10$)	1 (0.6%)	11 (6.4%)
DES-SN5YR	High- z ($z_{\text{HD}} \geq 0.10$)	17 (1.3%)	33 (2.4%)
Pantheon+	Low- z ($z_{\text{HD}} < 0.15$)	100 (15.6%)	39 (6.1%)
Pantheon+	High- z ($z_{\text{HD}} \geq 0.15$)	20 (3.1%)	75 (11.8%)

vs original masses

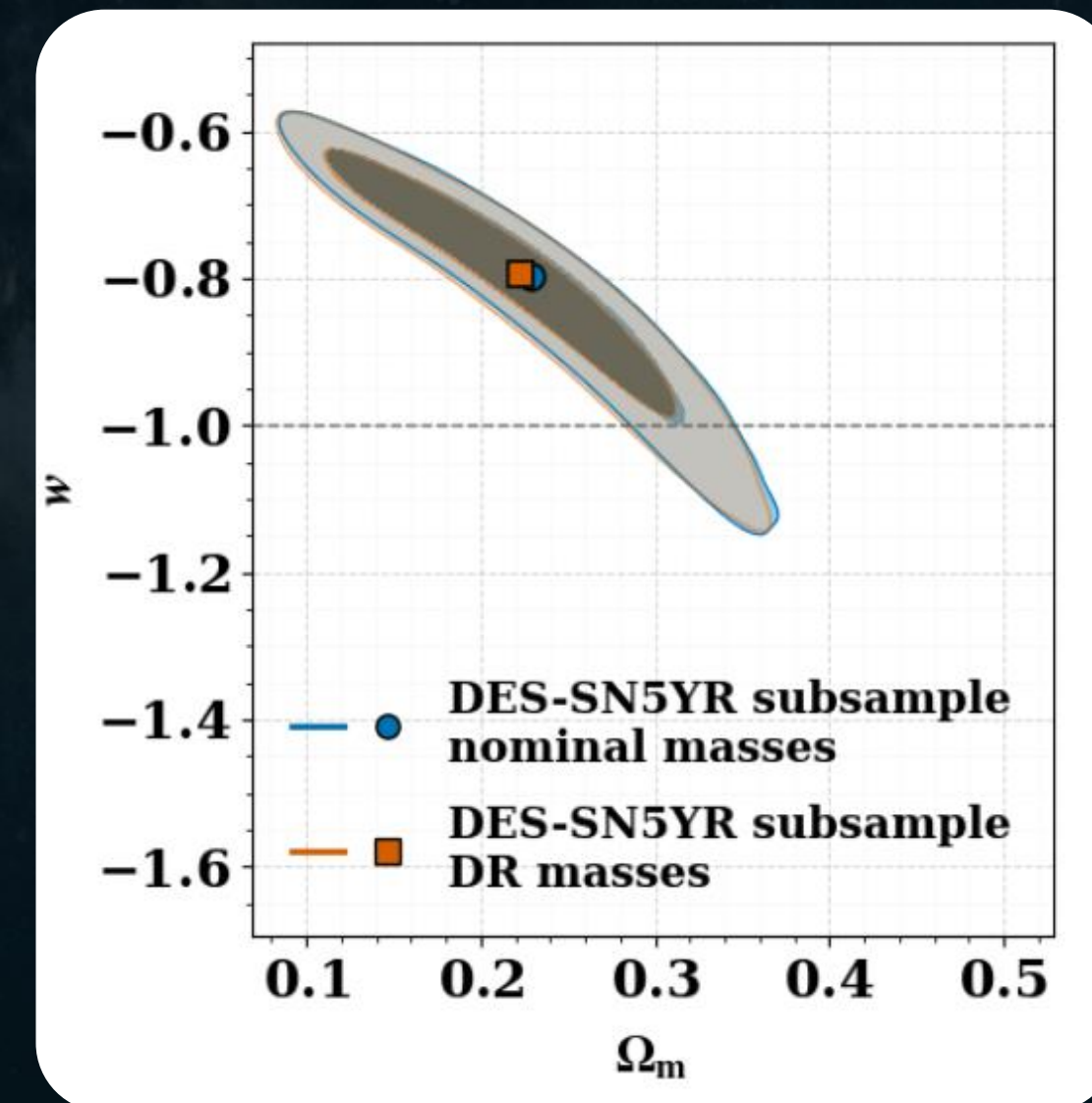


HOST MASS IMPACT

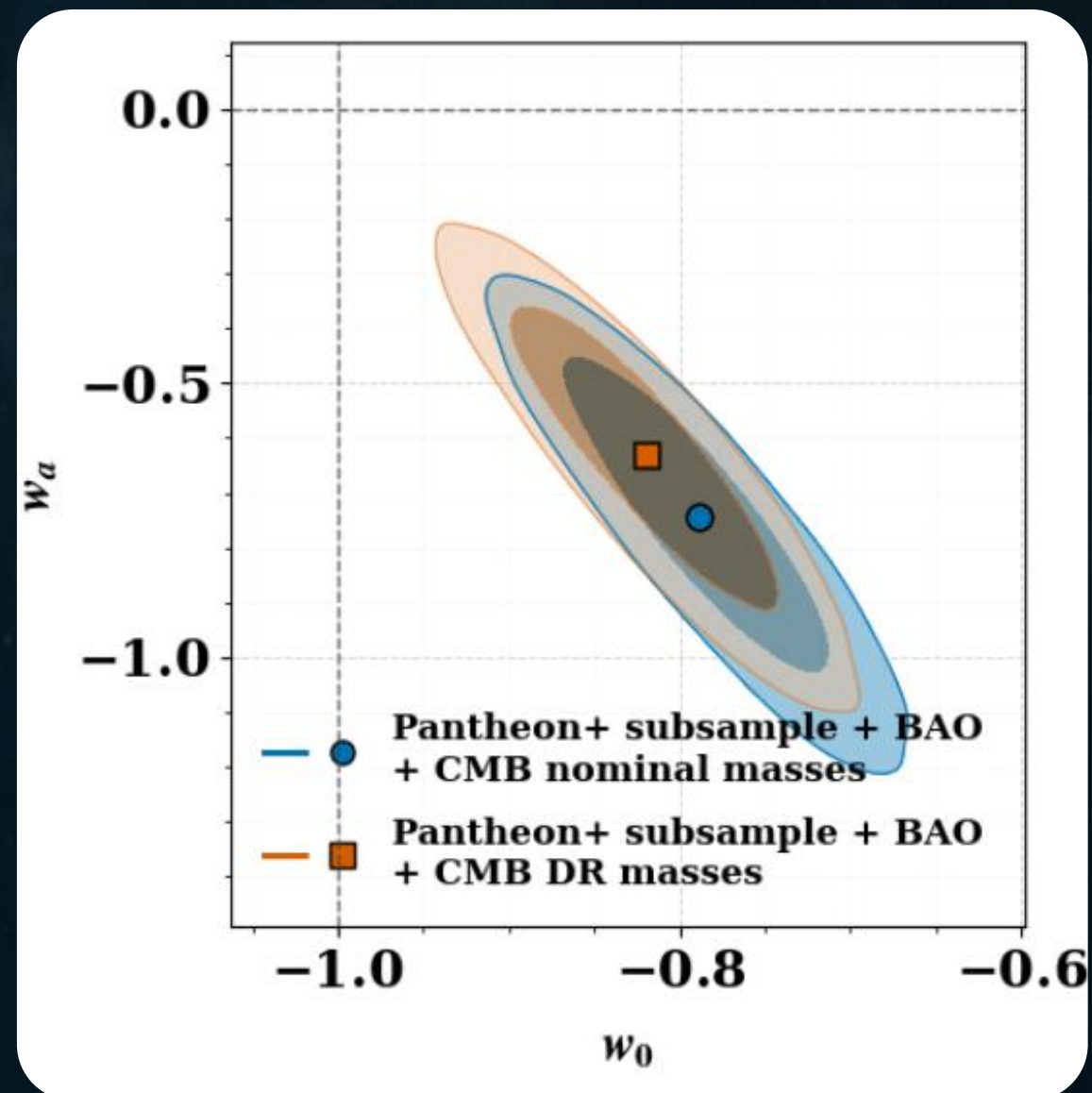
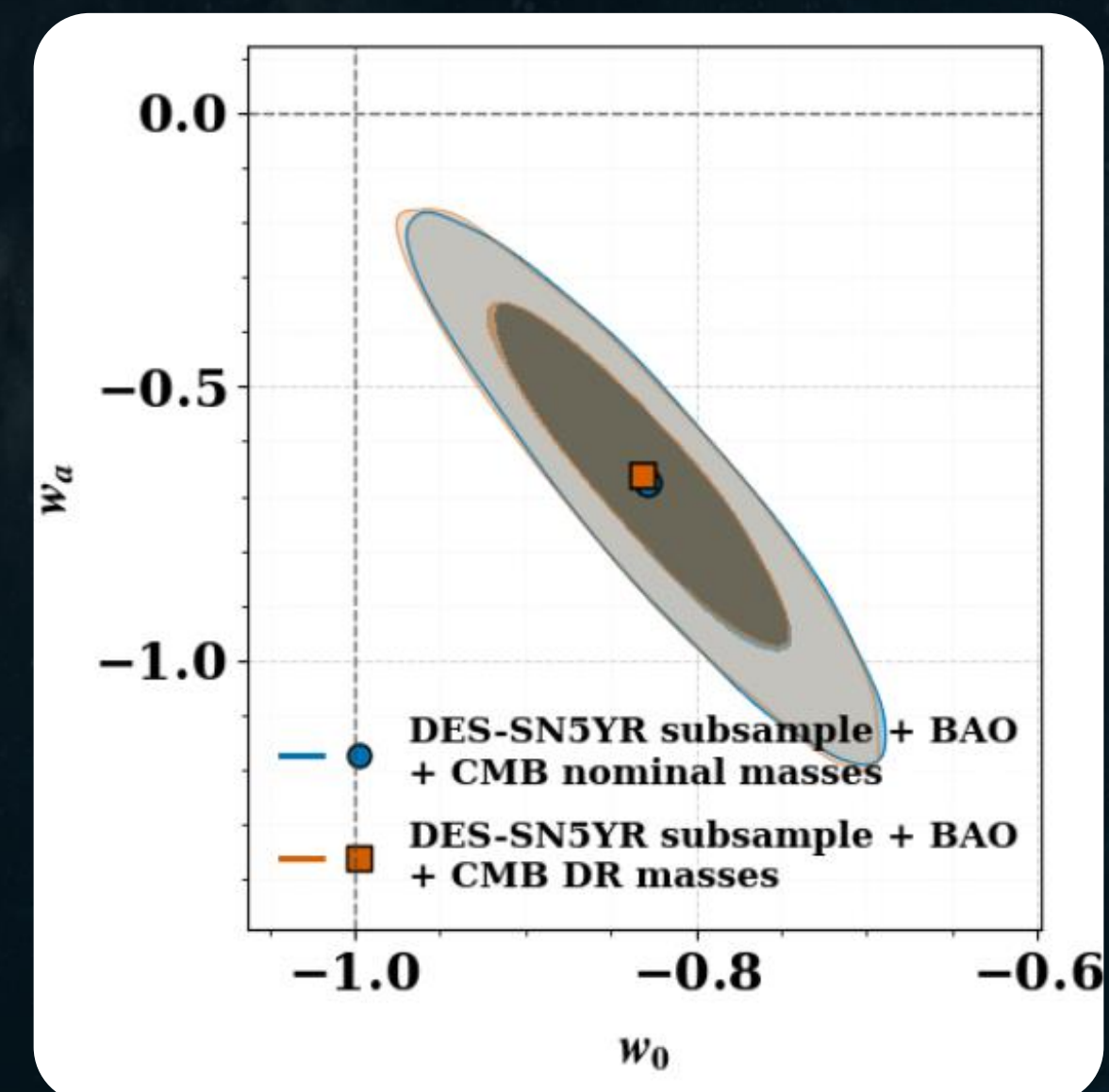
Pantheon+



DES-SN5YR



HOST MASS IMPACT

Pantheon+**DES-SN5YR**

Ryan Camilleri

On behalf of the Unite team!

COSMOLOGY RESULTS

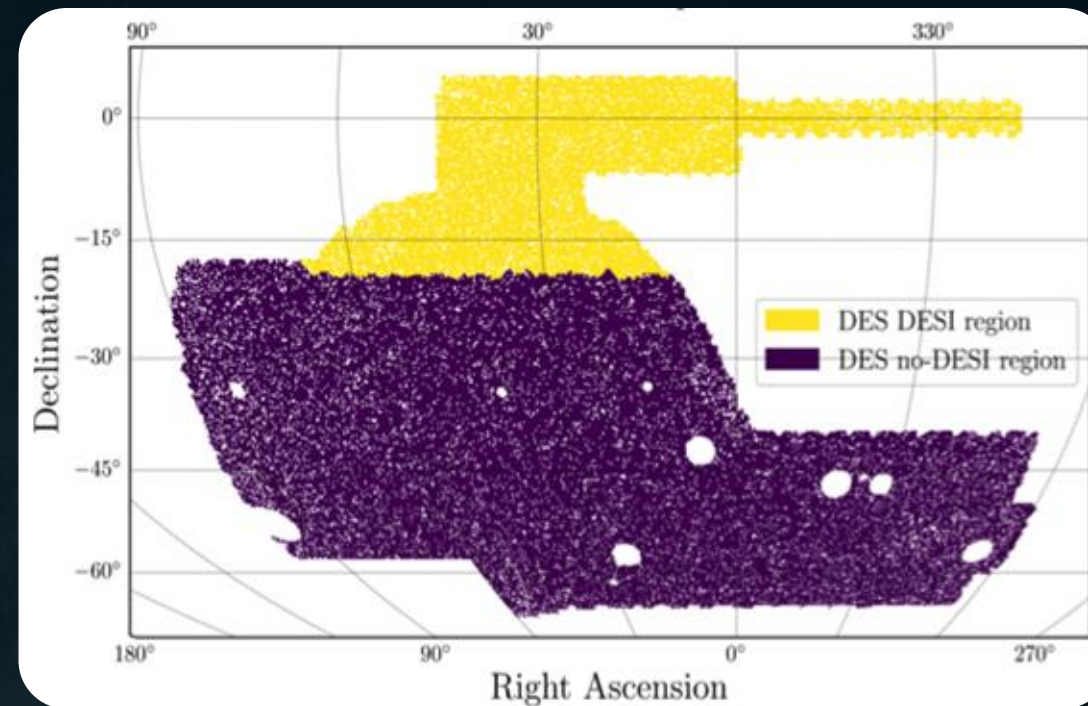
Camilleri et al: 2609.05053
Lee et al.: 2609.05321

Data URL



DES & DESI BAO

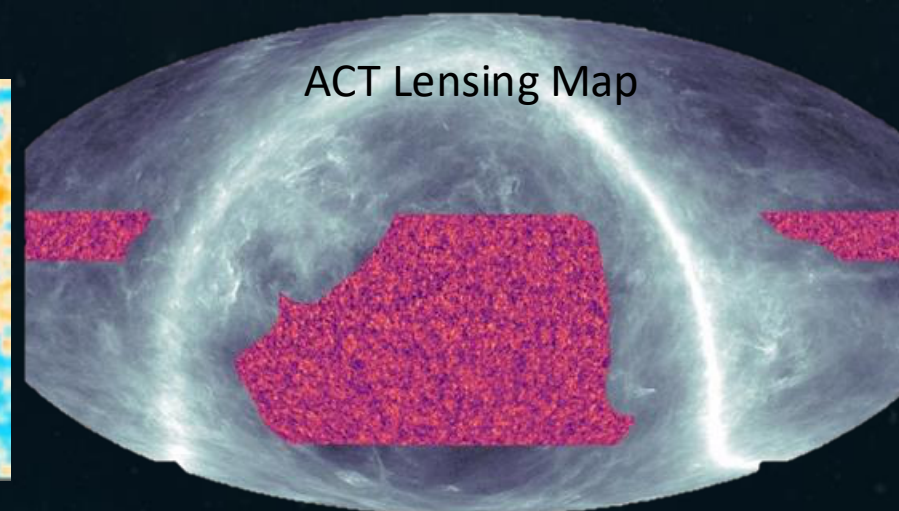
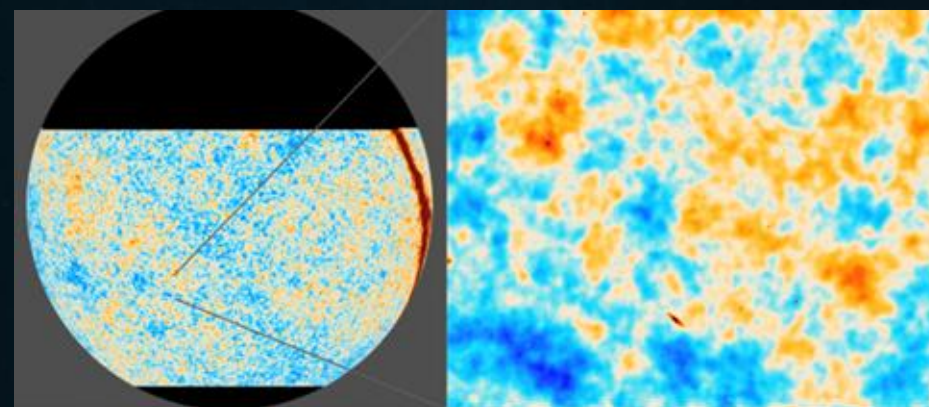
DES Collaboration 2024
DESI Collaboration 2025
Mena-Fernández et al. 2025



EXTERNAL PROBES

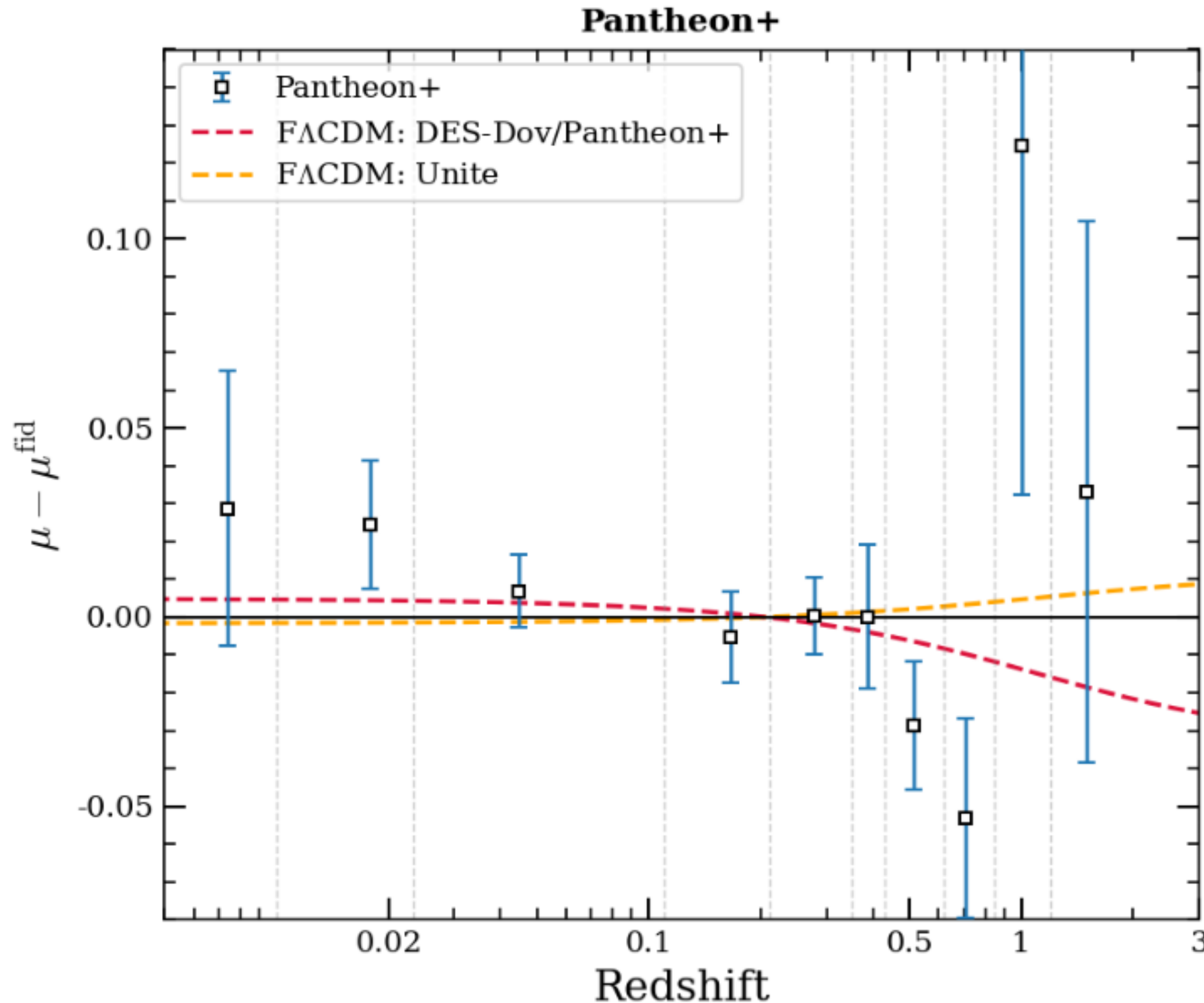
Planck & ACT & SPT CMB

Planck Collaboration 2020
Atacama Cosmology Telescope; Louis et al. 2025
South Pole Telescope; Camphius et al. 2026



HUBBLE DIAGRAM CHANGES

Camilleri et al: 2609.05053

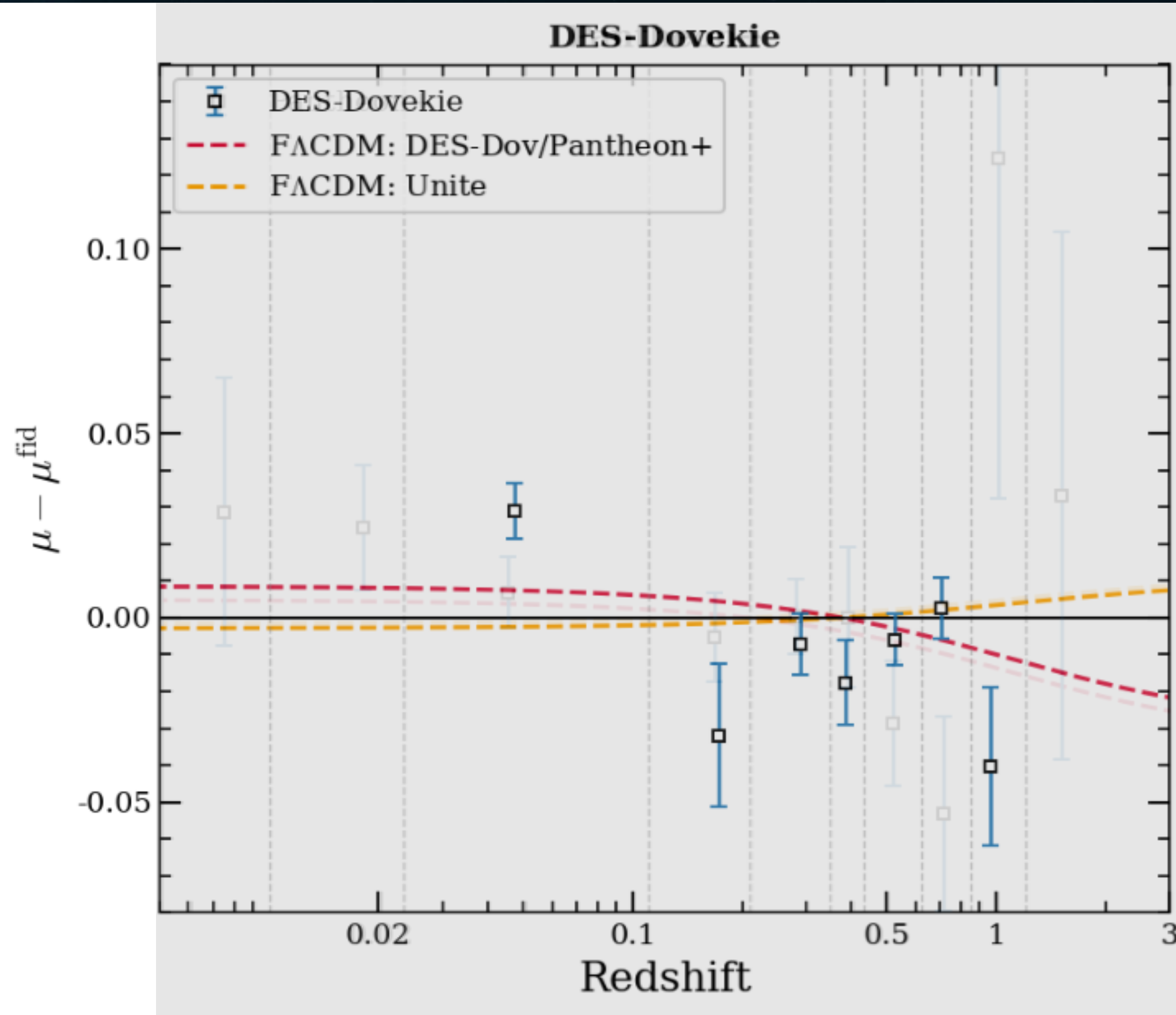


Binning is only for visualization

Bins show interesting regions

HUBBLE DIAGRAM CHANGES

Camilleri et al: 2609.05053

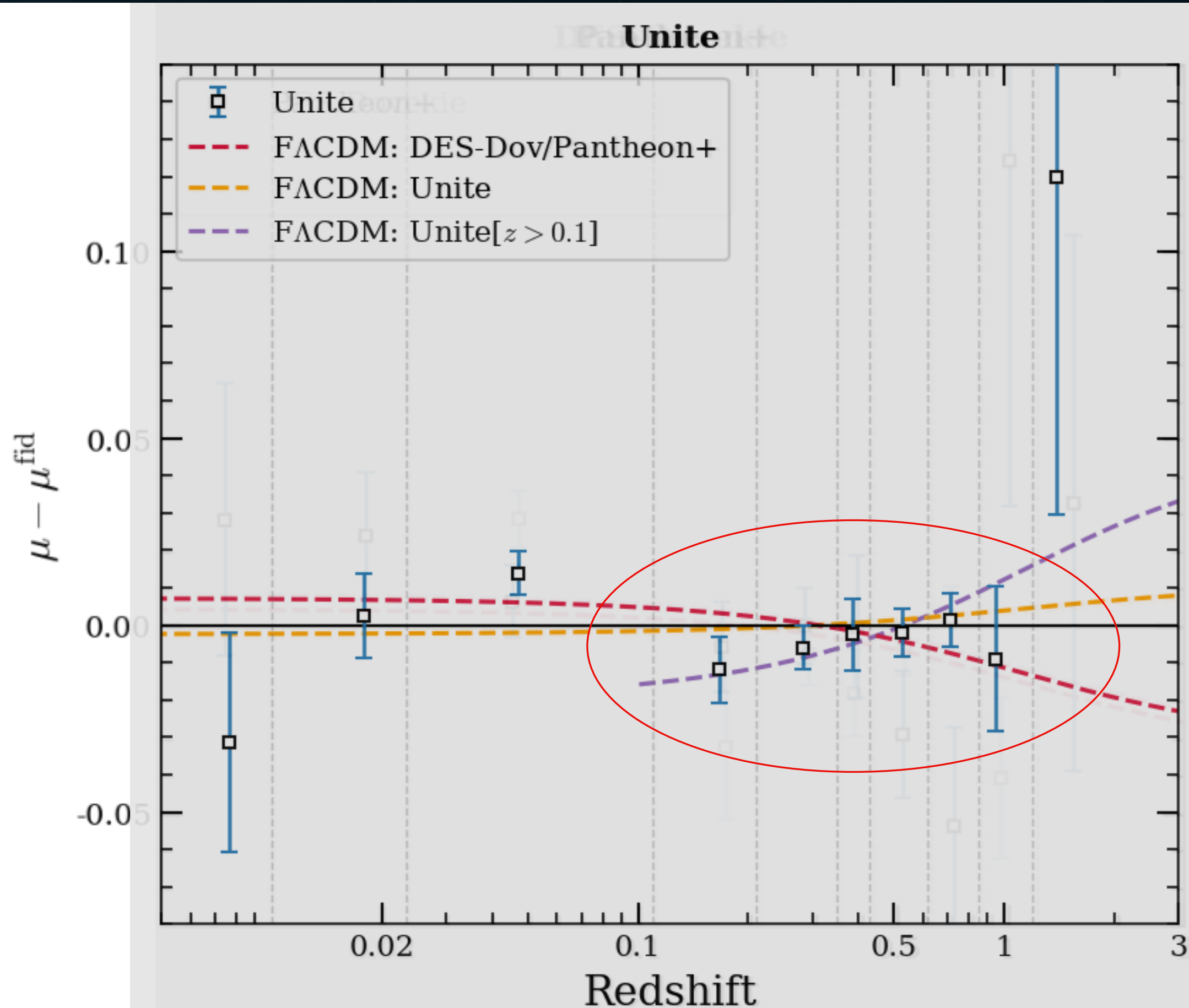


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Bins show interesting regions

HUBBLE DIAGRAM CHANGES

Camilleri et al: 2609.05053



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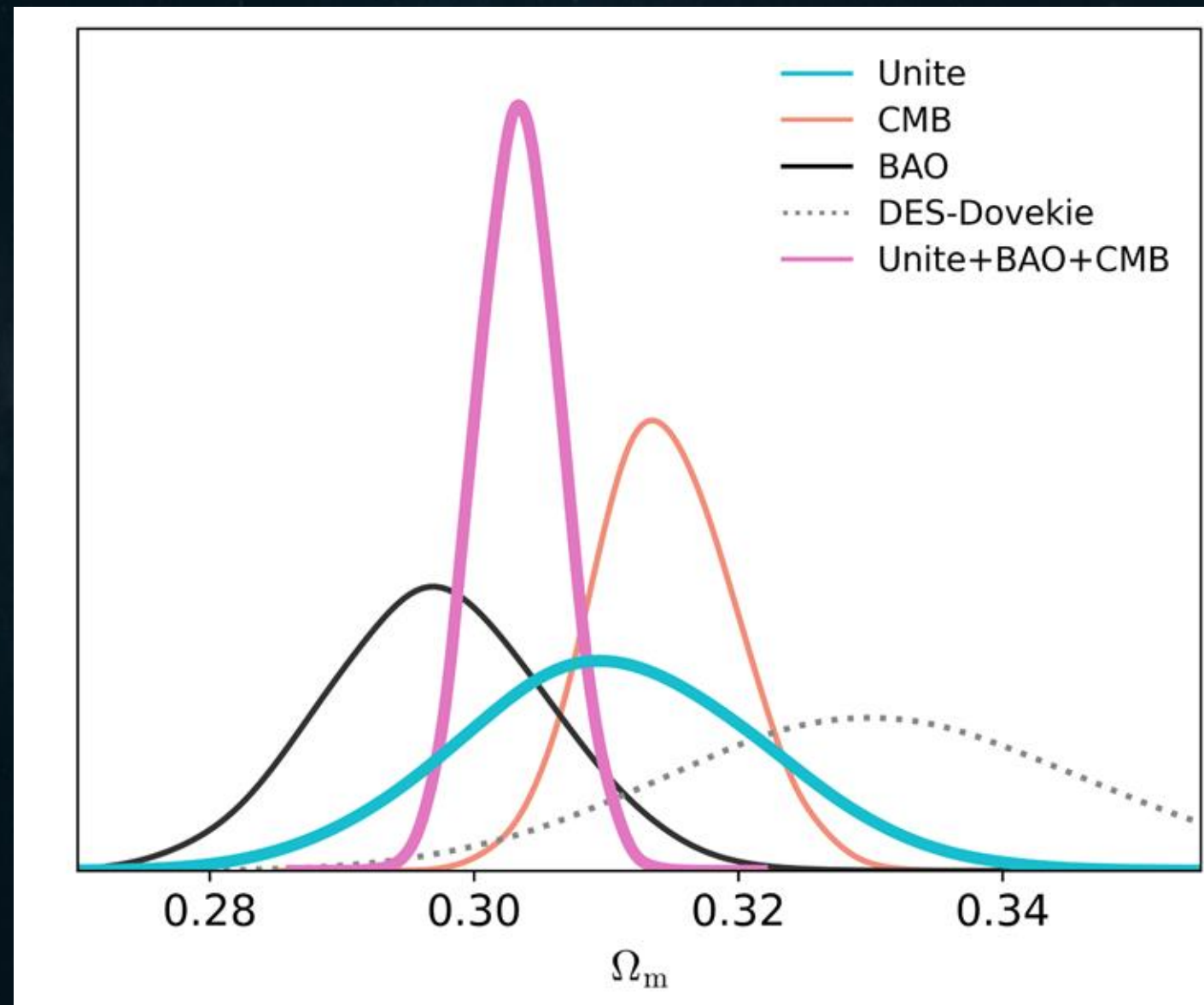
Bins show interesting regions

COSMOLOGY RESULTS

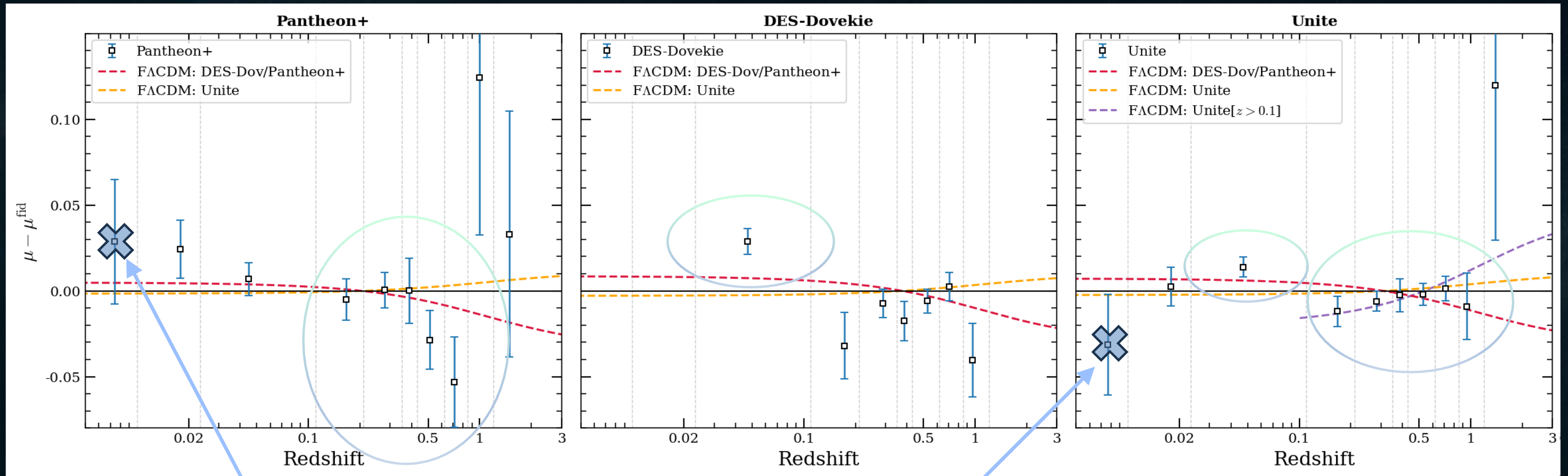
01**Flat- Λ CDM**

Unite alone: $\Omega_M = 0.310^{+0.012}_{-0.011}$

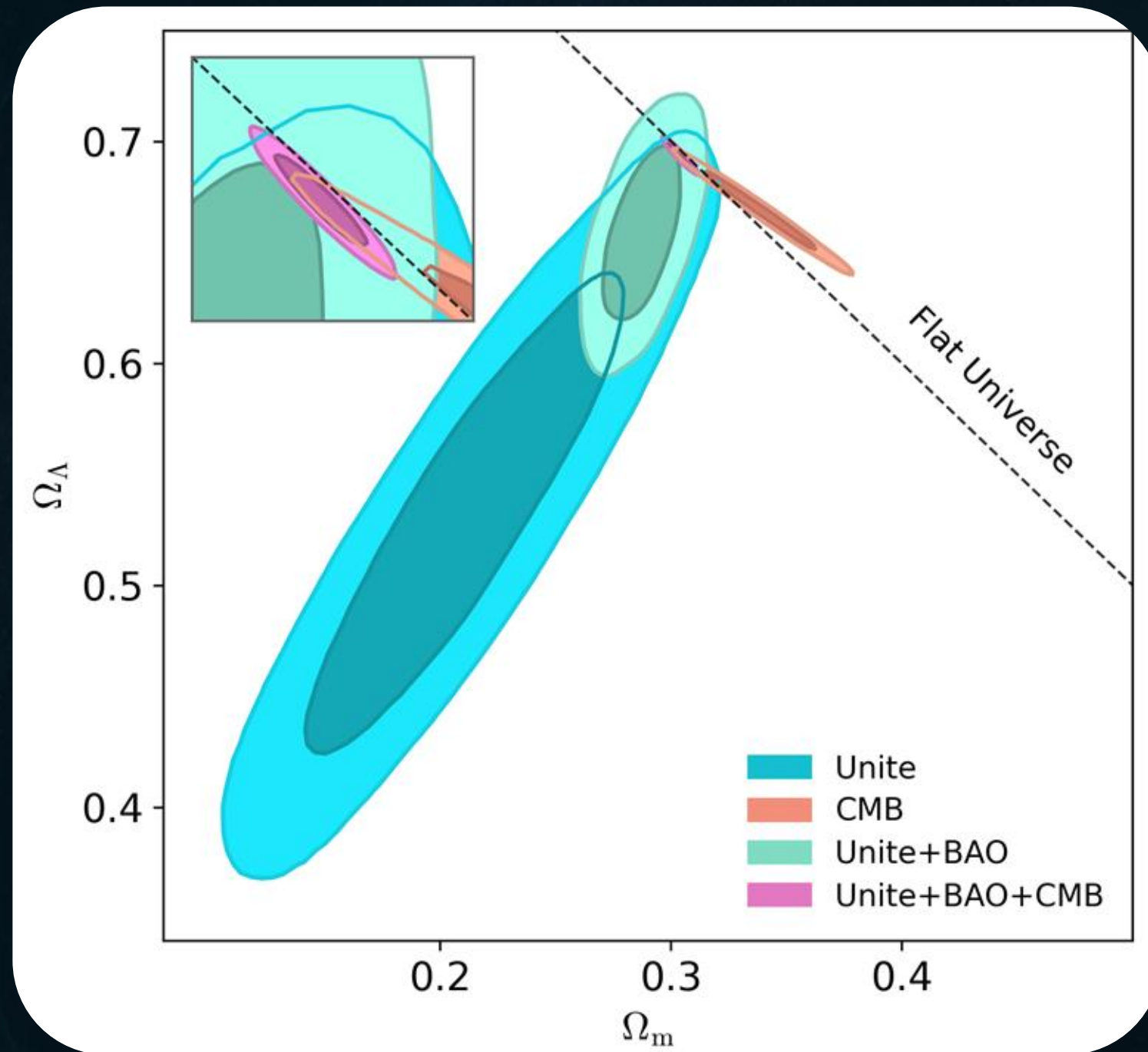
Unite + BAO + CMB: $\Omega_M = 0.303 \pm 0.003$



HUBBLE DIAGRAM CHANGES



Apparent shift is because the mean z of the bins differ slightly (magnitudes are actually very similar).



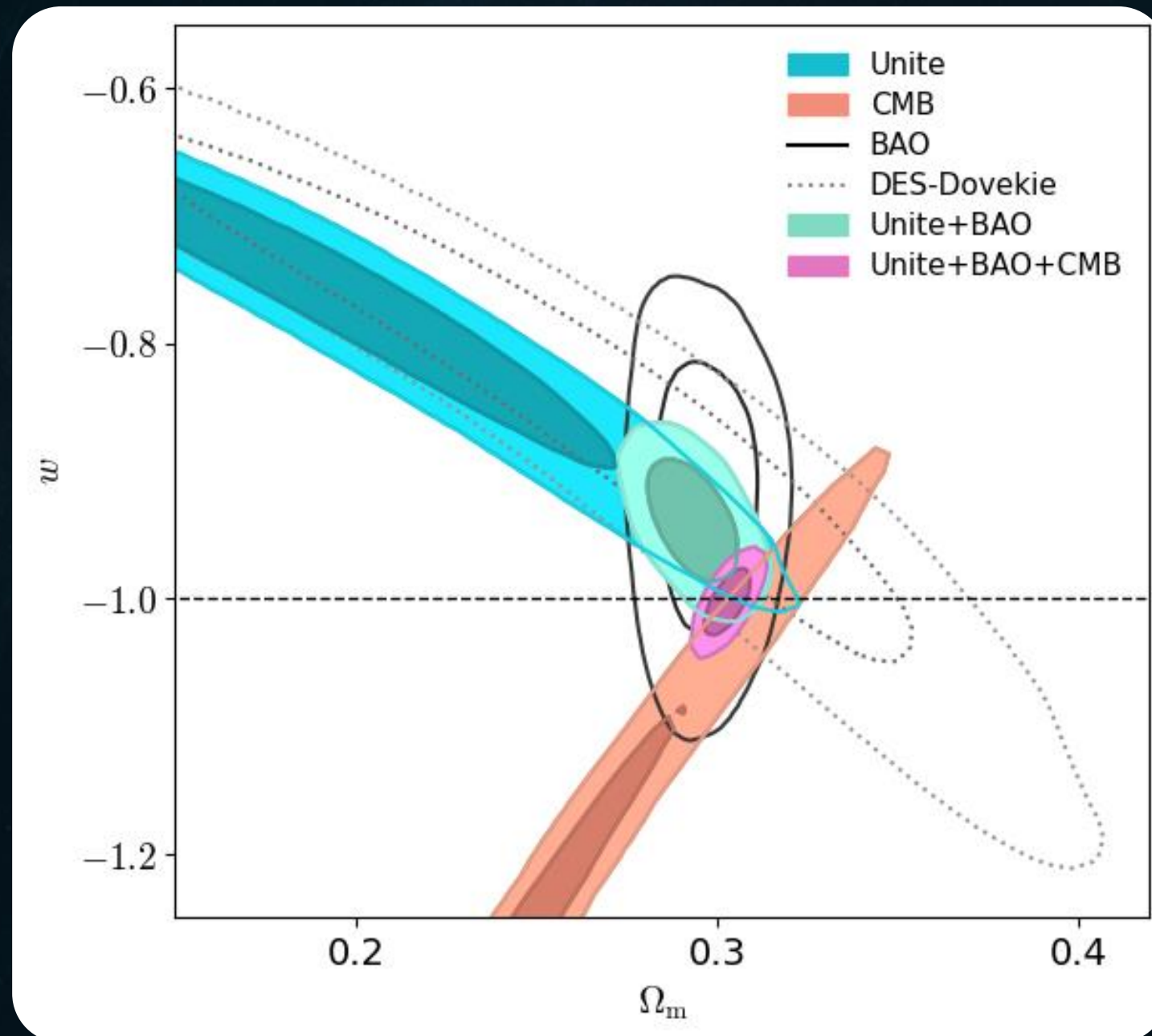
COSMOLOGY RESULTS

02**LCDM**

Unite alone: $\Omega_K = 0.26 \pm 0.11$

Unite + BAO + CMB: $\Omega_K = 0.003 \pm 0.001$

COSMOLOGY RESULTS



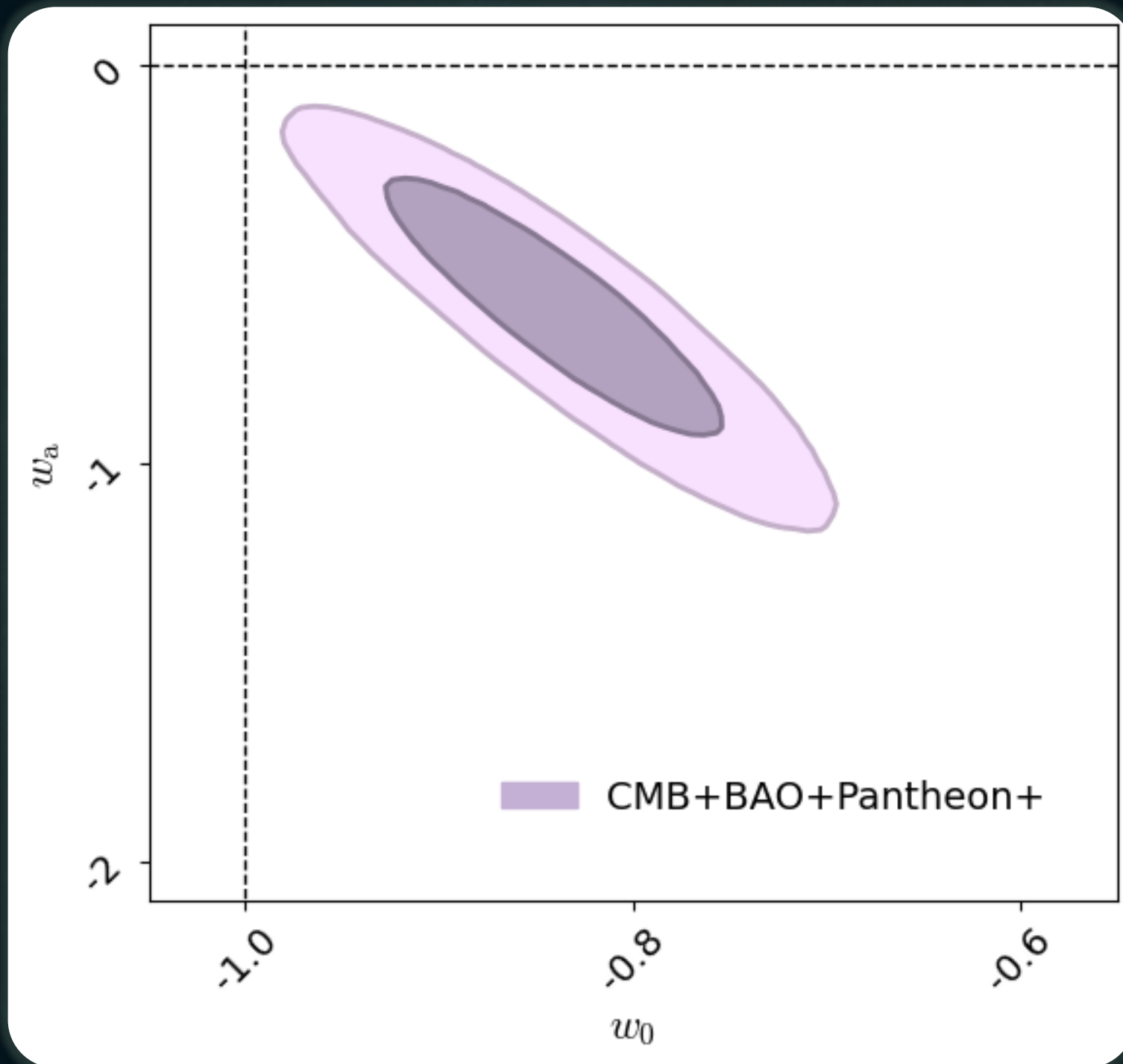
03

Flat- w CDM

Unite alone: $w = -0.764^{+0.078}_{-0.096}$

Unite + BAO + CMB: $w = -1.002 \pm 0.017^{**}$

**** We find a strong tension between Unite+BAO and CMB**



04

Flat- w_0w_a CDM

Unite + BAO + CMB:

$$w_0 = -0.861^{+0.044}_{-0.042}$$

$$w_a = -0.60^{+0.17}_{-0.19}$$

Tightest constraints to-date

 3.3σ from Flat- Λ CDM (MAP)

30% reduction in uncertainty of combined constraints

Bayesian evidence indicates a *weak* preference over Flat- Λ CDM

COSMOLOGY RESULTS

04

Flat- w_0w_a CDM

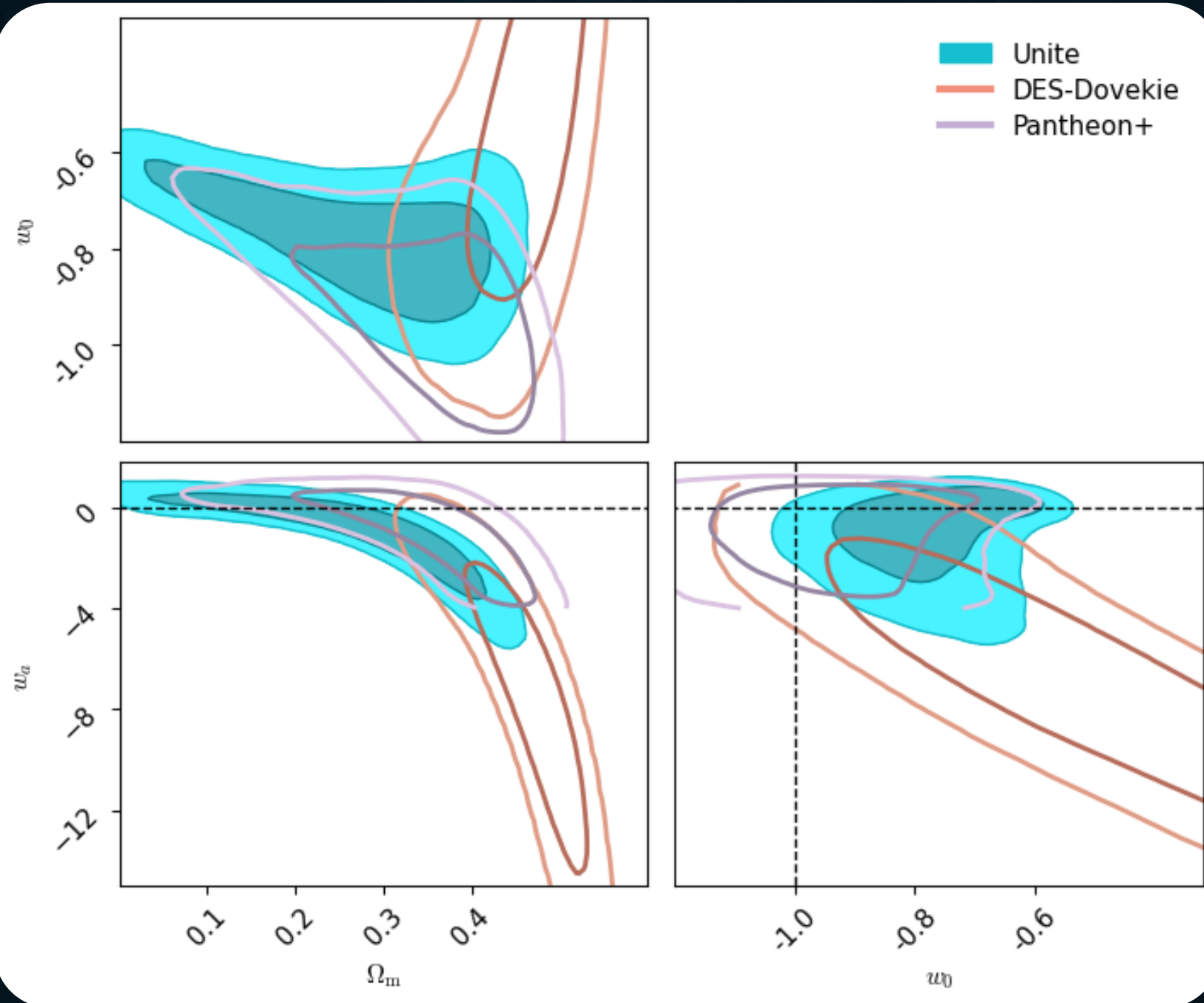
Unite alone:

$$w_0 = -0.779^{+0.099}_{-0.098}$$

$$w_a = -0.86^{+1.01}_{-1.78}$$

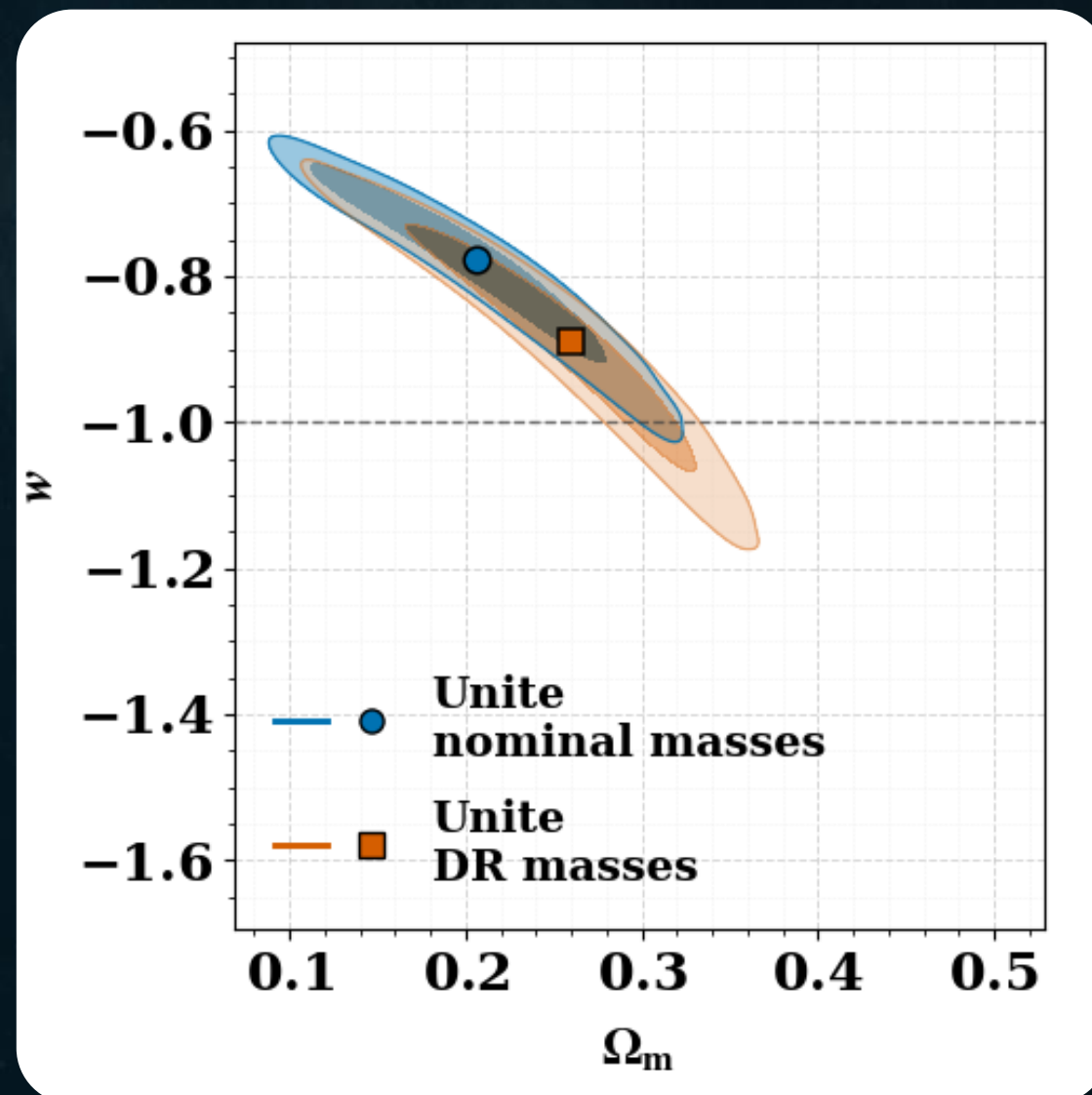
Supernovae consistent with $w_a = 0$

Contours shifted up degeneracy direction

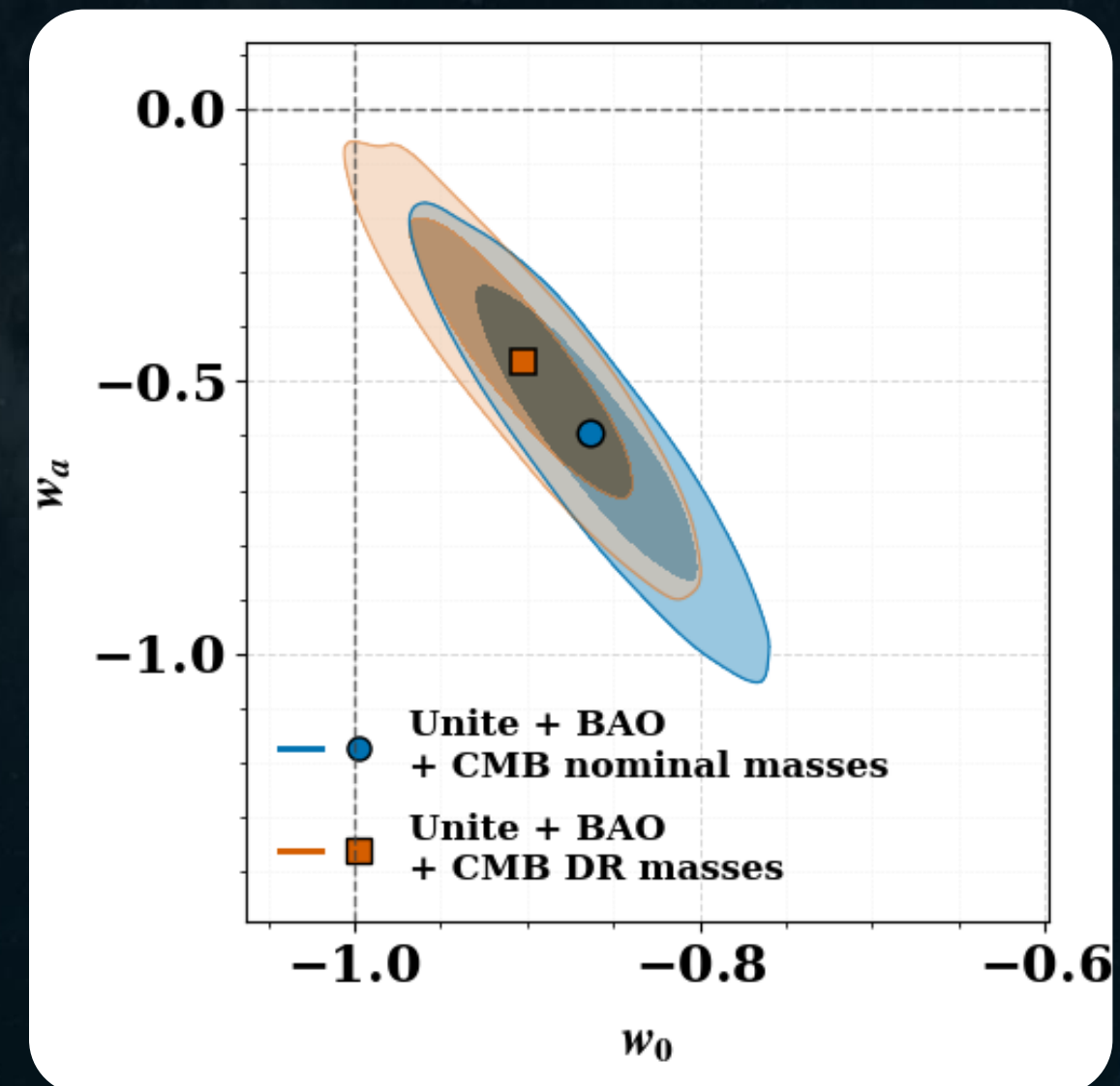


HOST MASS IMPACT

Unite, w CDM



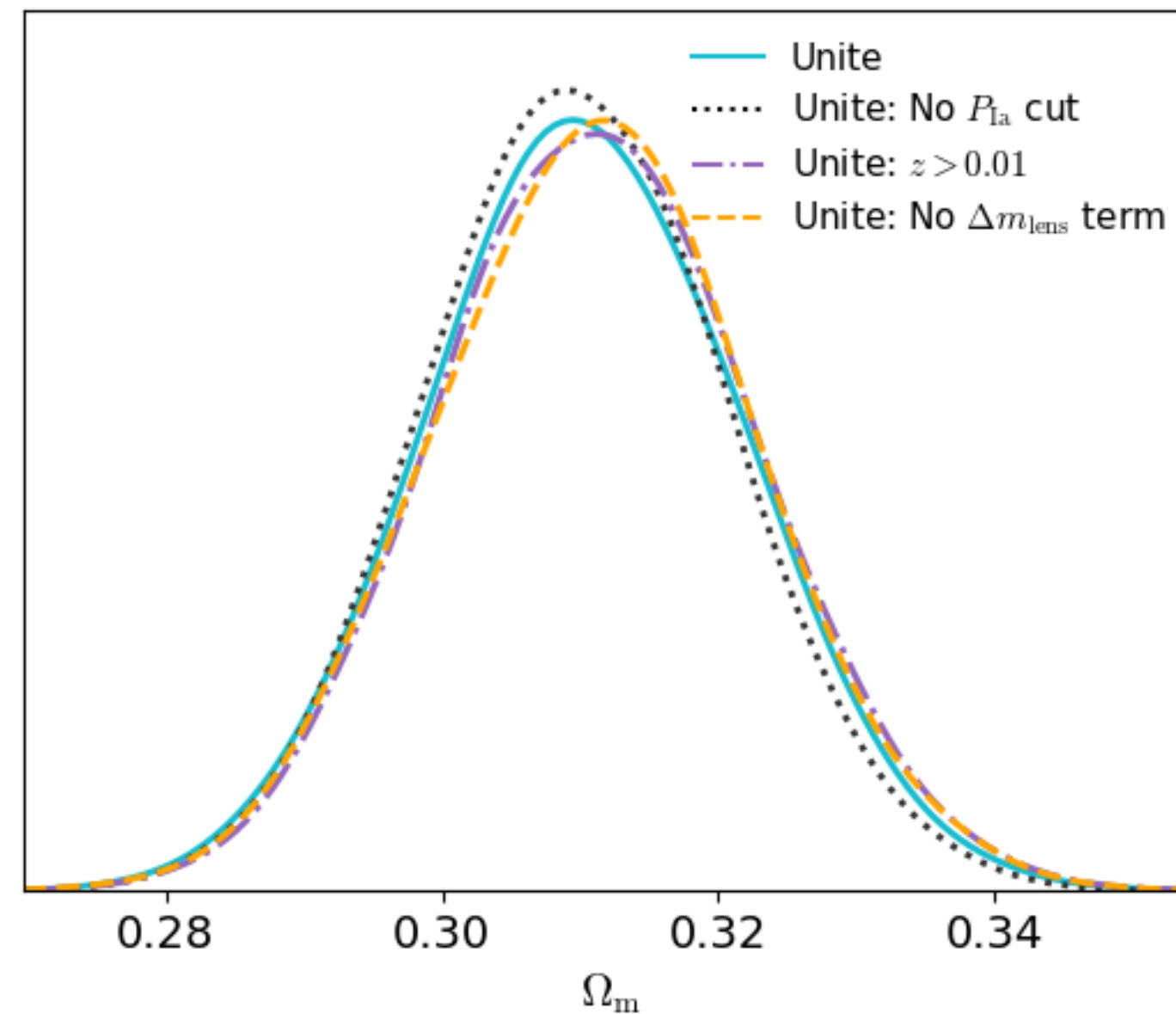
Unite, w_0w_a CDM



DIFFERENT ANALYSIS CHOICES

Camilleri et al: 2609.05053

Negligible impact on cosmology



1. Probability of being a type Ia cut

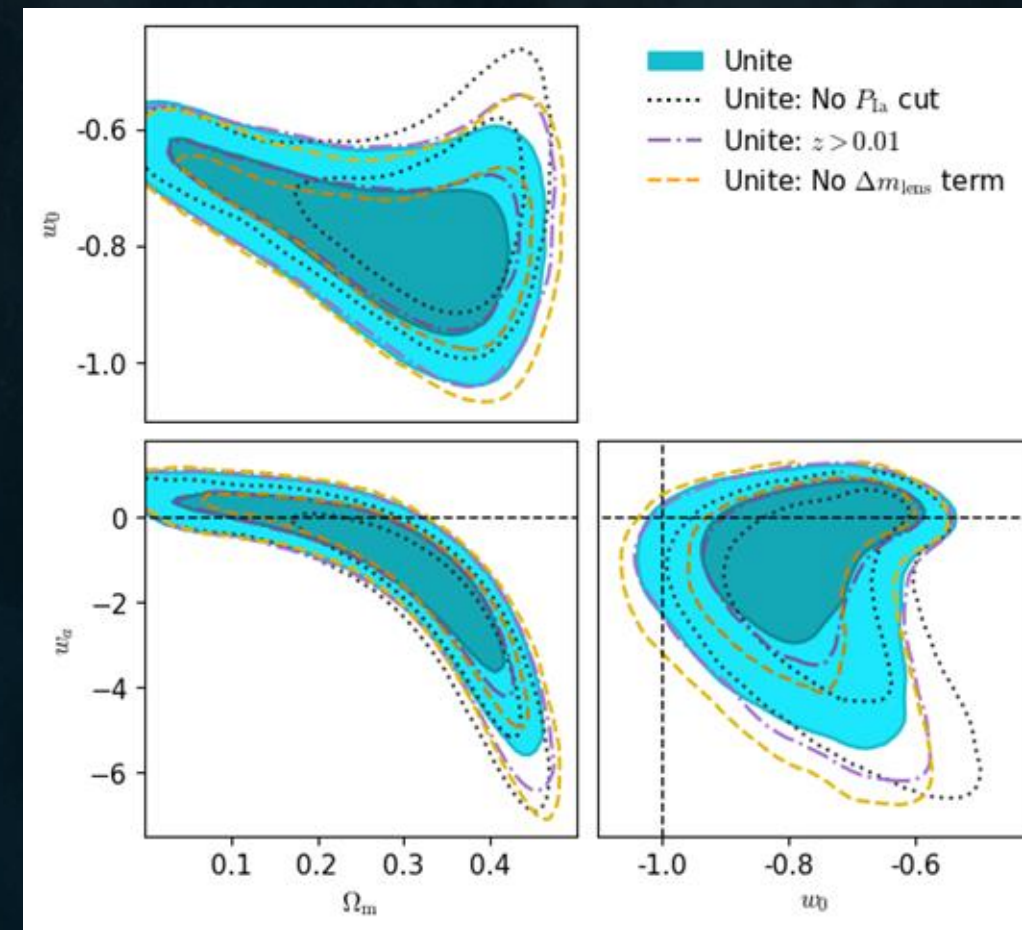
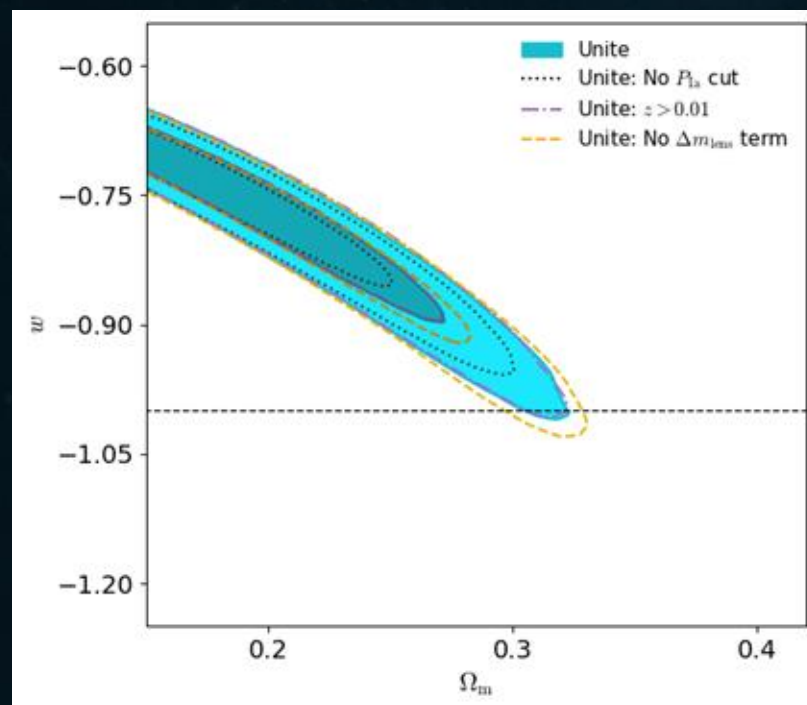
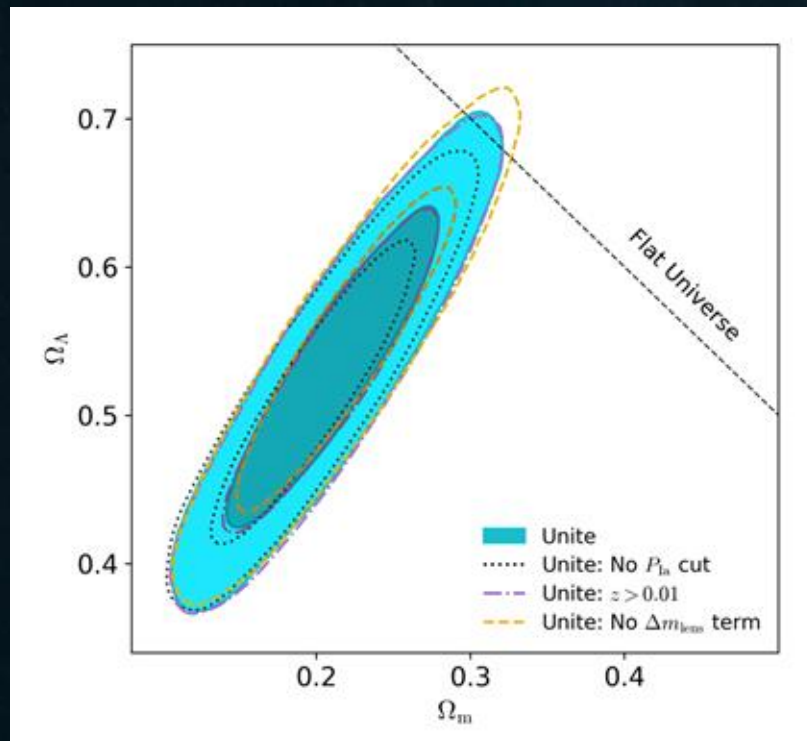
2. Making a low-redshift cut at $z > 0.01$

3. Removing the lensing correction

DIFFERENT ANALYSIS CHOICES

Camilleri et al: 2609.05053

Negligible impact on cosmology



1. Probability of being a type Ia cut

2. Making a low-redshift cut at $z > 0.01$

3. Removing the lensing correction

LOW-REDSHIFT SAMPLE

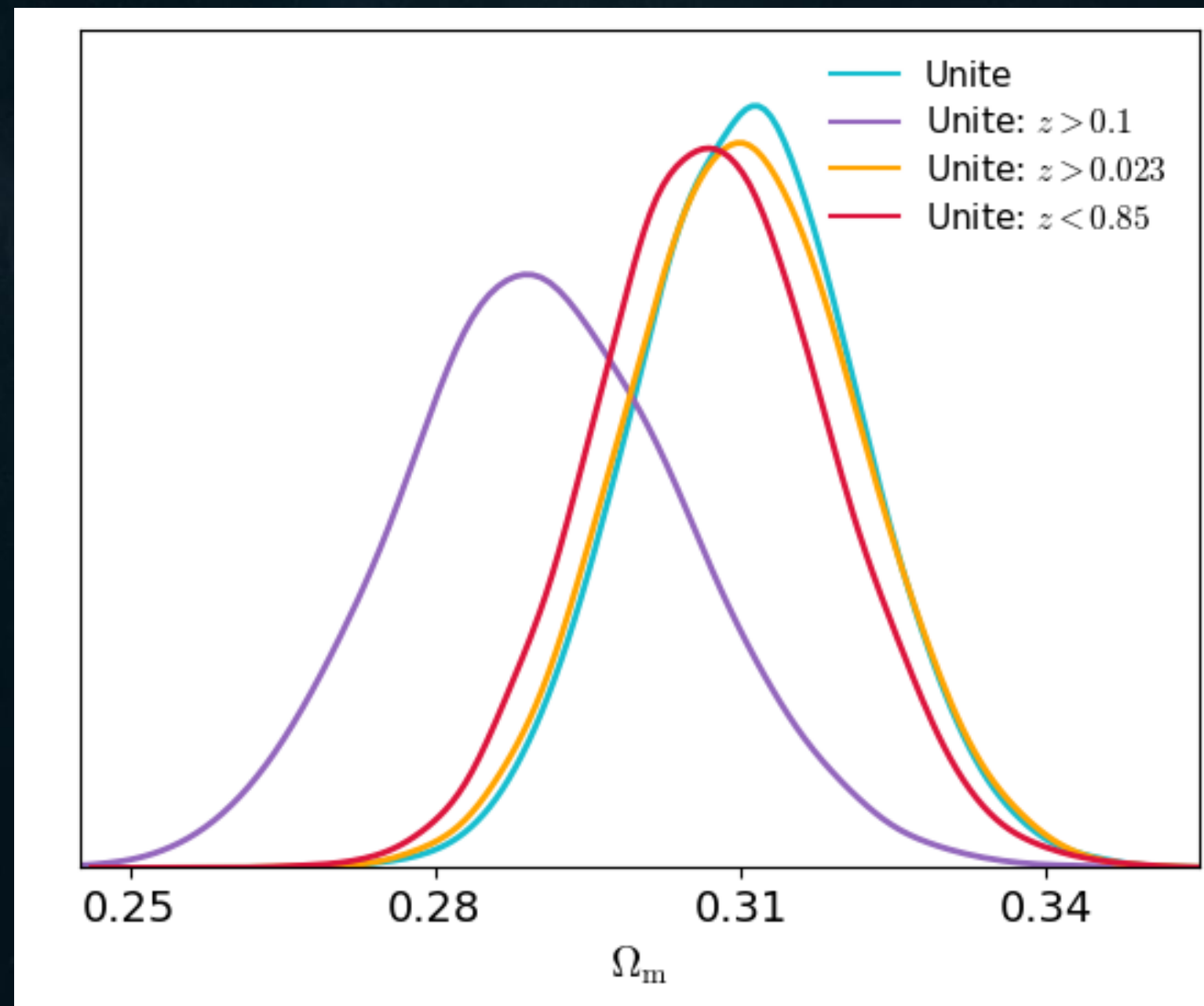
Camilleri et al: 2609.05053

Non-negligible impact on cosmology

Generally, our fits are insensitive to redshift cuts

Removing the entire low redshift sample moves our best fit by $\sim 2\sigma$

Underscores the importance of next-generation low- z samples

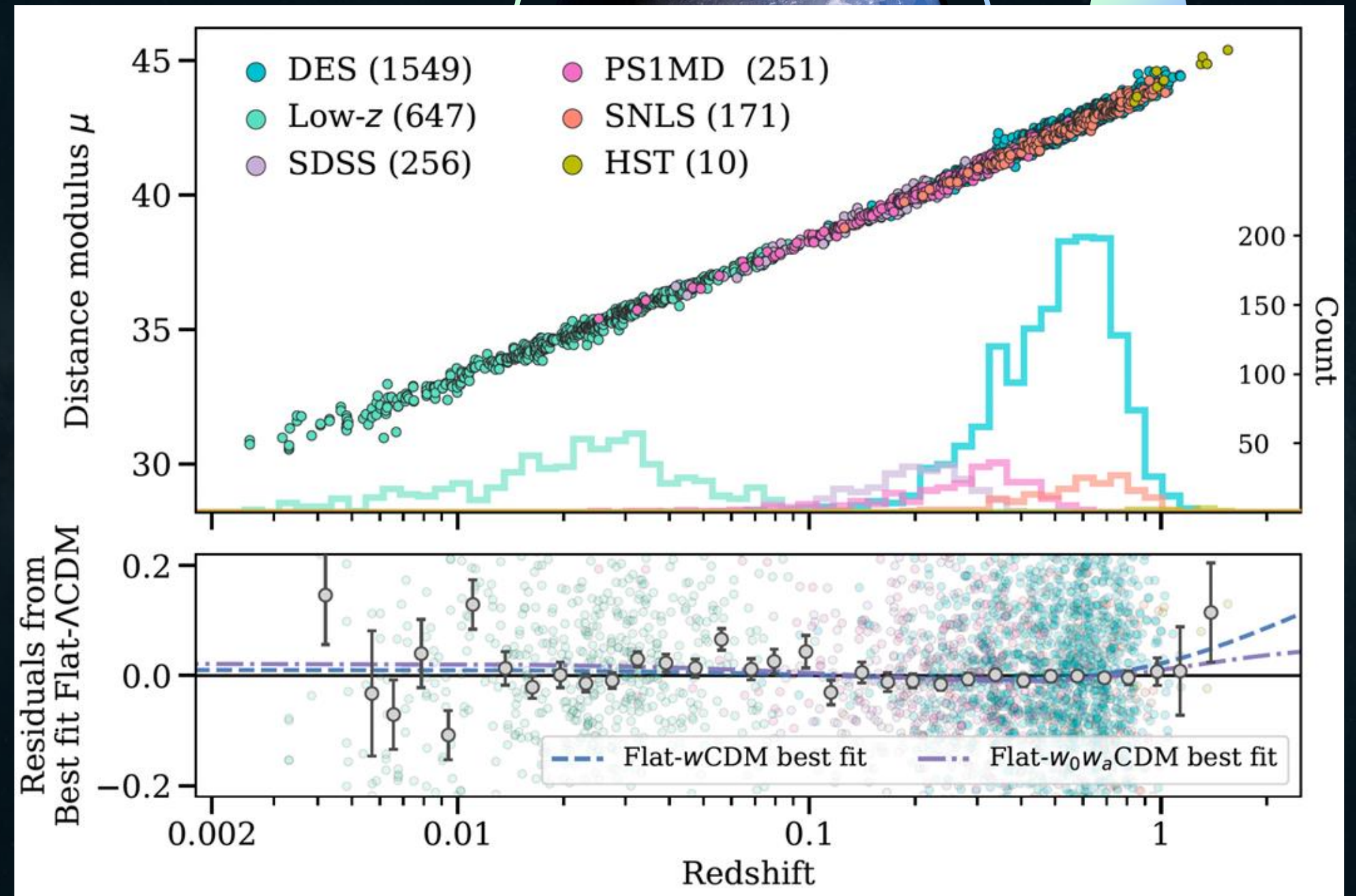


WHERE TO NEXT?

Constrain H_0

New data incoming!!

Apply UNITY to Unite



Unite is not a compilation we are trying to finish.

It is something we want the community to keep building.

Tamara Davis
Ryan Camilleri

SN - U N I T E !

Data release coming soon

THANK
YOU

